13.

Pipe & Cistern

Type - 1

1. A jug has two holes. First hole empties the jug in 15 minutes. Second hole empties the jug in 20 minutes. If the leakage is taking place at a constant rate, then in how much time the jug will be emptied by the both holes ?

(a) $\frac{4}{7}$ (b) $7\frac{4}{7}$

(c) $8\frac{5}{7}$ (d) $8\frac{4}{7}$

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (d) Time taken to empty the jug by both holes.

$=\frac{1}{15}+\frac{1}{20} \quad =\frac{4+3}{60}=\frac{7}{60}$

Required time $=\frac{60}{7}=8\frac{4}{7}$ minute

2. Two pipes can fill a tank in 20 hrs and 30 hrs respectively. If both the pipes are opened, then in how much time the tank will be fill ?

(a) 10 Hours (b) 12 Hours

(c) 18 Hours (d) 15 Hours

RRB JE - 26/06/2019 (Shift-I)

Ans : (b)

filled part in 1 hour by first pipe $=\frac{1}{20}$

and filled part in 1 hour by second pipe $=\frac{1}{30}$

filled part in 1 hour of both pipes

$$=\left(\frac{1}{20}+\frac{1}{30}\right)=\left(\frac{3+2}{60}\right)=\left(\frac{5}{60}\right)=\frac{1}{12} \text{ part}$$

So cistern will fill in 12 hours.

3. A tank can be filled by two pipes together in 45/4 minutes. Big pipe can fill the tank in 12 minute less as compared to small pipe. How much time will be taken by the big pipe alone to fill the tank ?

(a) 30 minute (b) 12 minute

(c) 18 minute (d) 24 minute

RRB RPF SI – 05/01/2019 (Shift-II)

Ans. (c) : Let time taken to fill the tank by big pipe = t

Time taken to fill the tank by small pipe = t + 12

As per the question,

$$\frac{1}{t}+\frac{1}{t+12}=\frac{1}{\frac{45}{4}}$$

$$\frac{t+12+t}{t(t+12)}=\frac{4}{45}$$

$$90t+540=4t^2+48t$$

$$4t^2-42t-540=0$$

$$2t^2-21t-270=0$$

$$2t(t-18)+15(t-18)=0$$

$$(t-18)(2t+15)=0$$

$t - 18 = 0$

$t = 18$ minute

4. Two pipes A and B can fill a tank in 45 hrs and 36 hrs respectively. If both the pipes are opened together, then how long will it take to fill the tank ?

(a) 10 Hours (b) 20 Hours

(c) 2 Hours (d) 5 Hours

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans : (b)

filled part by tank A in one hour $=\frac{1}{45}$

filled part by tank B in one hour $=\frac{1}{36}$

filled part by both (A+B) in one hour

$$=\frac{1}{45}+\frac{1}{36}=\frac{4+5}{180}=\frac{9}{180}=\frac{1}{20}$$

So all tank will fill by both pipes in 20 hours.

5. If two flood gates A and B work together then the reservoir will be filled in 6 hours. Gate A fills the reservoir 5 hour faster than Gate B. The Fast flood gate A will fill the reservoir in how many hours ?

(a) 5 Hours (b) 10 Hours

(c) 7 Hours (d) 13 Hours

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (b) Suppose B reservoir fills = in (x+5) hours and A reservoir fills = in x hours

As per the question,

$$\frac{1}{x}+\frac{1}{x+5}=\frac{1}{6}$$

$$\frac{x+5+x}{x^2+5x}=\frac{1}{6}$$

$12x + 30 = x^2 + 5x$

$x^2 - 7x - 30 = 0$

$x^2 - 10x + 3x - 30 = 0$

$(x - 10)(x+3) = 0$

$\boxed{x=10}$

Hence fast flood gate A will the reservoir in 10 hours.

6. Pipe A and B can fill a tank in 30 and 45 min. respectively, while pipe C can empty the whole tank in 60 minutes. If the three pipes are opened together, then how long will it take to fill an empty tank.

(a) $\left(34+\frac{1}{2}\right)$ minute (b) 60 minute

(c) $\left(18+\frac{5}{7}\right)$ minute (d) $\left(25+\frac{5}{7}\right)$ minute

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (d) Capacity of pipe A in 1 hour $=\frac{1}{30}$

capacity of pipe B in 1 hour $=\frac{1}{45}$

capacity of pipe C in 1 hour $=\frac{1}{60}$

filled part by three pipes in 1 hour $=\frac{1}{30}+\frac{1}{45}-\frac{1}{60}$

$=\frac{6+4-3}{180}$

$=\frac{7}{180}$

Required time = $=\frac{180}{7}=\left(25+\frac{5}{7}\right)$min

7. Pipe A can fill an empty pool in 14 hours. Together with pipe B it can fill the empty pool in 12 hours. So pipe B can fill the empty pool in ____ hours ?

(a) 84 (b) 75

(c) 78 (d) 77

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (a) Time taken to fill the pool by A =14 hours

filled part by A in 1 hour $=\frac{1}{14}$ part

Time taken to fill the pool by A and B together = 12 hours

filled part by A and B in 1 hour $=\frac{1}{12}$ part

Suppose B can fill pool in n hours then filled part by B in 1 hour $=\frac{1}{n}$ part

from the question–

So, $\frac{1}{14}+\frac{1}{n}=\frac{1}{12}$

$\frac{1}{n}=\frac{1}{12}-\frac{1}{14}$

$\frac{1}{n}=\frac{7-6}{84}\Rightarrow\frac{1}{n}=\frac{1}{84}\Rightarrow\boxed{n=84}$

Hence time taken to fill the pool by B = 84 hours

8. A pipe can fill a tank in 12 hrs and the second pipe can fill it in 15 hrs. If both the pipes are opened at the same time, then how much time will it take to fill half of the tank ?

(a) $4\frac{2}{3}$ Hours (b) $3\frac{1}{3}$ Hours

(c) $6\frac{2}{3}$ Hours (d) $2\frac{1}{3}$ Hours

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (b) Filled part of cistern by pipe A in 1 hour $=\frac{1}{12}$ part

filled part of cistern by pipe B in 1 hour $=\frac{1}{15}$ part

Suppose half of the tank will be filled by both pipes in t hours.

As per the question,

$\Rightarrow \frac{t}{12}+\frac{t}{15}=\frac{1}{2}$

$\Rightarrow \frac{5t+4t}{60}=\frac{1}{2}$

$\Rightarrow 9t=30$

$\Rightarrow t=\frac{10}{3}$

$\Rightarrow t=3\frac{1}{3}$

Hence half part of cistern will fill in $3\frac{1}{3}$ hours.

9. Pipe J and K can fill a tank in 15 and 20 minutes respectively. If both the pipes are opened together then, what time will be taken to fill the tank ?

(a) $17\frac{1}{2}$ minute (b) $11\frac{3}{5}$ minute

(c) $15\frac{2}{3}$ minute (d) $8\frac{4}{7}$ minute

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (d) Filled part by pipe J in1 minute = $\frac{1}{15}$ part

Filled part by pipe K in 1 minute = $\frac{1}{20}$ part

Filled part by pipe J and K in 1 minute = $\frac{1}{15}+\frac{1}{20}$

$=\frac{4+3}{60}=\frac{7}{60}$ part

Time taken to fill the whole tank = $\frac{60}{7}=8\frac{4}{7}$ minute

10. Pipe A and B can fill an empty tank in 10 and 15 hrs respectively. Both can fill the tank in ___ hrs.

(a) 6:10 (b) 6

(c) 4 (d) 6:15

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (b) Filled part of tank by pipe A in 1 hour $= \frac{1}{10}$ Part

Filled part of tank by pipe B in 1 hour $= \frac{1}{15}$ Part

Filled part of tank by pipe A and B in 1 hour $= \frac{1}{10} + \frac{1}{15}$

$= \frac{5}{30} = \frac{1}{6}$ Part

Hence time taken to fill the tank together = 6 hours

11. There are two pipes X and Y to fill an empty tank. Pipe X alone can fill 2/3 part of an empty tank in 10 hrs. Pipe Y alone can fill 1/6 part of an empty tank in 5 hrs. Both pipes following their own relative rate of flow will fill the tank in how many hours ?

(a) 30 Hours (b) 10 Hours
(c) 24 Hours (d) 15 Hours

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans. (b) :

Time taken in filling $\frac{2}{3}$ part by pipe X = 10 hours

$\therefore$ Time taken in filling all part by pipe X = $10 \times \frac{3}{2}$

= 15 hours

Time taken in filling $\frac{1}{6}$ part by pipe Y = 5 hours

$\therefore$ Time taken in filling all part by Y = 5 × 6 = 30 hours

Filled part of cistern by both pipe X and Y together in 1 hour

$= \frac{1}{30} + \frac{1}{15} = \frac{1+2}{30} = \frac{3}{30} = \frac{1}{10}$

$\therefore$ Time taken in filling whole tank = 10 hours

12. 44 pipes can fill a big tank of water in 21 hours. How much hours will be taken by 55 pipes to fill 5 such tanks ?

(a) 63 (b) 84
(c) 72 (d) 54

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (b) As per the question,

$$\frac{m_1 d_1}{w_1} = \frac{m_2 d_2}{w_2}$$

$$\frac{44 \times 21}{1} = \frac{55 \times x}{5}$$

$$\frac{44 \times 21}{1} = 11x$$

x = 4 × 21

x = 84 hours

13. Pipe A can fill an empty tank in 14 hours with another pipe B together A and B can fill the same tank in 10 hours. Pipe B alone can fill this empty tank in how many hours ?

(a) 35 (b) 20
(c) 30 (d) 25

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (a) According to the question,

Pipe A fills the empty cistern in 1 hour $= \frac{1}{14}$ Part

This empty cistern fills by (A+B) in 1 hour $= \frac{1}{10}$ Part

Time taken to fill the tank by pipe B $= \frac{1}{10} - \frac{1}{14}$

$= \frac{14-10}{140}$

$= \frac{4}{140} = \frac{1}{35}$ Part

Hence pipe B will fill the tank in 35 hours.

14. Pipe A and pipe B can fill a tank in 4 and 16 hours respectively. Both can fill the tank in how many hours ?

(a) $\frac{4}{15}$ Hours (b) $\frac{17}{3}$ Hours
(c) $\frac{16}{5}$ Hours (d) $\frac{16}{7}$ Hours

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (c) Filled part by pipe A in 1 hour = $\frac{1}{4}$ Part

Filled part by pipe B in 1 hour = $\frac{1}{16}$ Part

Filled part by pipe A and B in 1 hour

$= \frac{1}{4} + \frac{1}{16} = \frac{4+1}{16} = \frac{5}{16}$ Part

Hence time taken to fill the whole tank = $\frac{1}{5/16}$

$= \frac{16}{5}$ Hours

15. Two pipes A and B can fill a tank in X minute and 6 minute respectively. If both the pipes are working together, then they fill the tank in 1.5 minute. Find the value of X ?

(a) 1 minute (b) 2 minute
(c) 4 minute (d) 5 minute

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (b)

Time taken by pipe 'A' in filling the tank = X minute

Time taken by pipe 'B' in filling the tank = 6 minute

Total time taken by both in filling the tank = 1.5 minute

According to the question,

$$\frac{1}{X} + \frac{1}{6} = \frac{1}{1.5}$$

$$\Rightarrow \frac{6+X}{6X} = \frac{1}{1.5}$$

$$\Rightarrow 1.5(6+X) = 6X$$

$\Rightarrow 9+1.5X = 6X$

$\Rightarrow 6x - 1.5X = 9$

$\Rightarrow 4.5X = 9$

$\Rightarrow X = \frac{9}{4.5}$

X = 2 minute

16. There are two holes in a water tank. Hole 1 Alone takes 9 minutes to empty the tank and hole 2 alone can empty the tank in 6 minutes. If the leakage occurs at a fixed rate then by working the both pipes the tank will be emptied in how many minutes ?

(a) $3\frac{3}{5}$ (b) $\frac{3}{5}$

(c) $3\frac{1}{5}$ (d) $3\frac{2}{5}$

RRB NTPC 17.01.2017 Shift-1

Ans : (a) Portion emptied by hole- 1 in 1 minute $= \frac{1}{9}$

portion emptied by hole-2 in 1 minute $= \frac{1}{6}$

portion emptied by hole-1 and hole-2 of tank in 1 minute

$= \frac{1}{9} + \frac{1}{6} = \frac{9+6}{9\times 6} = \frac{15}{54}$

time taken by both hole together in emptying tank =

$= \frac{54}{15} = \frac{18}{5} = 3\frac{3}{5}$ minute

17. A container has two holes. The first hole can empty the container individually in 15 minutes and the second hole can empty the container individually in 10 minutes. If the leakage occurs at a fixed rate in the container, then by the both holes together opening, the container will be emptied in how many minutes ?

(a) 6 (b) $\frac{1}{6}$

(c) $\frac{1}{7}$ (d) 7

RRB NTPC 17.01.2017 Shift-3

Ans : (a) Leakage by hole-1 in 1 minute $= \frac{1}{15}$ part

leakage by hole-2 in 1 minute $= \frac{1}{10}$ part

leakage by both holes in 1 minute =

$\frac{1}{15} + \frac{1}{10} = \frac{25}{150}$ part

Hence the time taken by hole-1 and holes 2 to empty the container $= \frac{150}{25} = 6$ minute

Hence hole-1 and hole-2 will empty the container together in 6 minutes.

18. A tube has two holes. One hole alone empties it in 29 minutes. The second hole alone empties the tube in 6 minutes. If the leakage is happening at a fixed rate, then together the both hole can empty the tank in how much time?

(a) $\frac{34}{35}$ (b) $3\frac{34}{35}$

(c) $4\frac{33}{35}$ (d) $4\frac{34}{35}$

RRB NTPC 11.04.2016 Shift : 3

Ans : (d)

Portion emptied by hole-1 in 1 minute $= \frac{1}{29}$

portion emptied by hole-2 in 1 minute $= \frac{1}{6}$

leakage by hole-1 and hole-2 in 1 minute

$= \frac{1}{29} + \frac{1}{6} = \frac{6+29}{174} = \frac{35}{174}$

time taken by hole-1 and hole-2 to empty whole tank

$= \frac{174}{35} = 4\frac{34}{35}$ minute

19. A tank has two holes. The first hole alone empties the tank in 21 minutes and the second hole alone empties the tank in 7 minutes. If the leakage is happening at a fixed rate then how much time will be taken by the both holes to empty the tank?

(a) $\frac{1}{4}$ (b) $4\frac{1}{4}$

(c) $5\frac{2}{4}$ (d) $5\frac{1}{4}$

RRB NTPC 07.04.2016 Shift : 1

Ans : (d) Time taken by holes in emptying the tank

$= \frac{1}{21} + \frac{1}{7} = \frac{1+3}{21} = \frac{4}{21}$

Required time $= \frac{21}{4} = 5\frac{1}{4}$ minute

20. There is two holes in a glass filled with water. On opening the first hole alone, this glass is emptied in 9 minutes and on opening the second hole alone then the glass is emptied in 3 minutes. If the leakage of the glass is happening regularly, then on opening both the holes, the glass will be emptied in how many minutes?

(a) $\frac{1}{4}$ (b) $2\frac{2}{4}$

(c) $3\frac{1}{4}$ (d) $2\frac{1}{4}$

RRB NTPC 19.01.2017 Shift : 1

Ans : (d) Time taken to empty the glass by two holes when two holes are opened together $= \frac{1}{9}+\frac{1}{3}=\frac{3+9}{9\times 3}$

$$=\frac{9\times 3}{12}=\frac{27}{12}=\frac{9}{4}=2\frac{1}{4}$$

21. There are two holes in a tank. First hole empties the tank in 3 minutes. The second hole empties the tank in 5 minutes. If the leakage is happening at a fixed rate, then how much time will be taken by the both holes to empty the tank?

(a) $\frac{7}{8}$ (b) $2\frac{7}{8}$

(c) $1\frac{5}{8}$ (d) $1\frac{7}{8}$

RRB NTPC 26.04.2016 Shift : 2

Ans : (d) The tank emptied in 1 minute by the first hole $=\frac{1}{3}$

The tank emptied in 1 minute by the second hole $=\frac{1}{5}$

$\therefore$ portion emptied by both holes in 1 minute

$$=\frac{1}{3}+\frac{1}{5}=\frac{5+3}{15}=\frac{8}{15}$$

Hence the entire tank will be empty in $\frac{15}{8}=1\frac{7}{8}$ minute

22. A bowl has two holes. First hole alone can empty the bowl in 4 minutes. The second hole alone can empty the bowl in 12 minutes. If the leakage is happening at a fixed rate, then both holes will empty the bowl in how much time?

(a) 1.33 (b) 3.5

(c) 4 (d) 3

RRB NTPC 30.04.2016 Shift : 3

Ans : (d) Time taken by first hole = 4 minute

time taken by second hole = 12 minute

time taken by both holes together $=\frac{1}{4}+\frac{1}{12}$

$$=\frac{3+1}{12}=\frac{4}{12}=\frac{1}{3}$$

Hence required time = 3 minute

23. Two pipes X and Y can individually fill a tank in 48 and 72 minutes, respectively. If they are opened simultaneously, how long will it take for the tank to fill?

(a) 39.4 minutes (b) 60 minutes

(c) 28.8 minutes (d) 24 minutes

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (c) : Filled part of the tank by pipes X and Y in 1 minute

$$=\frac{1}{48}+\frac{1}{72}=\frac{72+48}{48\times 72}=\frac{120}{3456}\text{ part}$$

Time taken by pipe X and Y in filling the whole tank

$$=\frac{3456}{120}=28.8\text{ minute}$$

Type - 1

24. A pipe can fill a tank in 20 minutes, but an outlet pipe can empty it in 28 minutes. When both the pipes are opened, then in how much time the tank will be filled?

(a) 72 minutes (b) 56 minutes

(c) 96 minutes (d) 70 minutes

RRB JE - 28/05/2019 (Shift-I)

Ans : (d) Portion filled by first pipe in 1 minute $=\frac{1}{20}$

portion emptied by out let pipe in 1 minute $=\frac{1}{28}$

portion filled in 1 minute by both pipe $=\frac{1}{20}-\frac{1}{28}$

$$=\frac{28-20}{20\times 28}=\frac{8}{20\times 28}=\frac{1}{70}\text{ part}$$

So time taken in filling the tank = 70 minute

25. Two pipes 'P' and 'Q' together can fill a tank in 4 hrs. When both the pipes are opened individually then Q takes 6 hours more as compared to P to fill the tank. P alone can fill the tank in how much time?

(a) 5 hours (b) 6 hours

(c) 8 hours (d) 7 hours

RRB JE - 02/06/2019 (Shift-I)

Ans : (b) Suppose pipe P takes x hours in filling cistern then Q will take time to fill cistern = x + 6

filled part by (P + Q) in 1 hour $=\frac{1}{4}$

filled part by P in 1 hour $=\frac{1}{x}$

filled part by Q in 1 hour $=\frac{1}{x+6}$

According to the question,

$$\frac{1}{x}+\frac{1}{x+6}=\frac{1}{4}$$

$$\frac{(x+6)+(x)}{x(x+6)}=\frac{1}{4}$$

$(2x + 6) \times 4 = x^2 + 6x$

$8x + 24 = x^2 + 6x$

$x^2 - 2x - 24 = 0$

$x^2 - (6-4)\,x - 24 = 0$

$(x^2 - 6x) + (4x - 24) = 0$

$x(x - 6) + 4\,(x - 6) = 0$

$(x + 4)\,(x - 6) = 0$

$x - 6 = 0,\ x = 6$

Hence P can fill tank in 6 hours

26. 2/5th part of tank A is filled with water. Pipe P can fill it in 10 minutes, while pipe Q can empty it in 6 minutes. If P and Q are opened together, then in how much time tank will be emptied?

(a) 6 minute (b) 6.5 minute
(c) 5.5 minute (d) 7 minute

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (a) 1 minute work of pipe P and Q $= \frac{1}{10} - \frac{1}{6}$

$= \frac{3-5}{30} = \frac{-2}{30} = \frac{-1}{15}$

time taken by P and Q in emptying the tank

$= \frac{\frac{2}{5}}{\frac{1}{15}} = \frac{2}{5} \times \frac{15}{1} = 6$ minute

27. A tank can be filled in 9 hours but due to a leakage, it takes 1 hour more. If tank is completely full, then due to leakage it will be empty in how much time?
(a) 60 hours (b) 75 hours
(c) 30 hours (d) 90 hours

RRB JE - 01/06/2019 (Shift-III)

Ans. (d) Filled part in 1 hour $= \frac{1}{9}$ part

leakage part in 1 hour $= \frac{1}{10}$ part

if tank is completely full

portion emptied by leakage in 1 hour $= \frac{1}{9} - \frac{1}{10}$

$= \frac{10-9}{90} = \frac{1}{90}$

Hence cistern will be empty by leakage in 90 hours.

28. 2/5 part of a water tank is filled. Pipe A can fill the tank in 12 minutes and pipe B can empty the same tank in 6 minute. If both pipes are opened together, then to fill the empty tank or to empty the full tank, what time will be taken?
(a) It will fill in 4.8 minutes.
(b) It will empty in 5.6 minutes.
(c) It will empty in 4.8 minutes.
(d) It will fill in 5.6 minutes.

RRB RPF SI – 10/01/2019 (Shift-II)

Ans. (c) Filled portion by pipe A in 1 minute $= \frac{1}{12}$

Emptied portion by pipe B in 1 minute $= \frac{1}{6}$

suppose time taken to fill tank completely or empty $= t$ minute

According to the question,

Time taken by both pipe to empty or fill the tank

$= \frac{t}{12} - \frac{t}{6} = \frac{2}{5}$

$\Rightarrow \frac{t-2t}{12} = \frac{2}{5}$

$\Rightarrow \frac{-t}{12} = \frac{2}{5}$

$\Rightarrow t = -4.8$ minute

where (–) indicates of emptying the cistern

29. A pipe can fill an empty cistern in 7.8 hours while another can drain the cistern when full in 19.5 hours. Both the pipes were turned on when the cistern was half-empty. How long will it take for the cistern to be full?
(a) 5.2 hours (b) 3.9 hours
(c) 7.8 hours (d) 6.5 hours

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (d) : Filled portion by first pipe in one hour $= \frac{1}{7.8}$

Emptied portion by second pipe in one hour $= \frac{1}{19.5}$

filled portion by both pipes in one hour

$= \frac{1}{7.8} - \frac{1}{19.5}$

$= \frac{19.5-7.8}{19.5 \times 7.8} = \frac{11.7}{152.1}$

time taken in filling whole cistern $= \frac{152.1}{11.7}$ hour

Now time taken in filling remaining half portion of tank

$= \frac{152.1}{11.7} \times \frac{1}{2}$

$= \frac{152.1}{23.4} = 6.5$ hours

30. A pipe can fill any tank in 7/4 hours while the other pipe can empty the full tank in 21/8 hours. Both pipes were opened at that time when the tank was 2/3 empty. How much time will be taken to fill the tank?
(a) 3 hours 20 minute (b) 3 hours 30 minute
(c) 3 hours 45 minute (d) 3 hours 15 minute

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (b) Suppose it will take t hours to fill the tank.

According to the question-

$\frac{t}{\frac{7}{4}} - \frac{t}{\frac{21}{8}} = \frac{2}{3}$

$\frac{4t}{7} - \frac{8t}{21} = \frac{2}{3}$

$\Rightarrow \frac{12t-8t}{21} = \frac{2}{3}$

$\Rightarrow \frac{4t}{21} = \frac{2}{3}$

$\Rightarrow t = \frac{2 \times 21}{3 \times 4}$

$\Rightarrow t = \frac{7}{2}$h

$\Rightarrow t = 3\frac{1}{2}$

$\Rightarrow t = 3 : \frac{1}{2} \times 60$

$\Rightarrow t = 3h : 30$ min

Hence it will take 3 hours 30 minutes to fill the tank.

31. A tap can fill the tank in 25 minutes and the other can empty the tank in 50 minutes. If both the taps are opened simultaneously, then what time will the tank be filled in time?

(a) 1 hour, 5 minutes (b) 50 minutes
(c) 55 minutes (d) 1 hour, 5 minutes

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (b) : Suppose filling tank = A
emptying tank = B

filled portion of tank by Tap A in 1 minute $= \frac{1}{25}$

emptying portion of tank by Tap B in 1 minute $= \frac{1}{50}$

Hence filled portion of cistern in 1 minute by both tap =

$$= \frac{1}{25} - \frac{1}{50} = \frac{2-1}{50} = \frac{1}{50}$$

Hence time taken in filling whole tank = 50 minute

32. A tank has two taps. One tap can fill the tank in 8 hours and the second tap can empty it in 10 hours. If the both taps were opened together, then what time will be taken to fill the tank?

(a) 40 (b) 20
(c) 30 (d) 50

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (a) : It takes 8 hours by X in filling the tank.
Where as it takes 10 hours by Y in emptying the tank.

∴ filled portion of tank in one hour

$$= \frac{1}{X} - \frac{1}{Y} = \frac{1}{8} - \frac{1}{10} = \frac{1}{40}$$

∴ time taken to fill the tank = 40 hours

33. A tank has two taps, first fills water in tank, and second empties the tank. Tank is filled in 4 hours, when only first tap is 12 hrs. opened and filled in 12 hours when both taps can opened. How much time is taken to empty the full tank when only second tap is opened?

(a) 10 hours (b) 4 hours
(c) 6 hours (d) 8 hours

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (c) Filled portion of tank in 1 hour by first tap

$$= \frac{1}{4} \text{ part}$$

filled portion of tank in 1 hour by both tap $= \frac{1}{12}$ part

∴ filled portion of tank in 1 hour by second tap

$$= \frac{1}{4} - \frac{1}{12} \text{ part}$$

$$= \frac{3-1}{12}$$

$$= \frac{2}{12} = \frac{1}{6}$$

Hence second tap will empty the filled cistern in 6 hours

34. Tank has two taps. Tap A is to fill the tank and tap B is to empty the tank. If only tap A can fill the tank in 35 hours, then only tap B can empty the full tank in 70 hours. Find the time taken to fill the half filled tank?

(a) 35 hours (b) 40 hours
(c) 70 hours (d) 55 hours

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (a) : Filled portion of cistern in 1 hour by tap A = $\frac{1}{35}$

emptied portion of cistern in 1 hour by tap B $= \frac{1}{70}$

suppose time taken in filling half empty cistern completely = t time

According to the question-

$$\frac{t}{35} - \frac{t}{70} = \frac{1}{2}$$

$$\frac{2t-t}{70} = \frac{1}{2}$$

$$\frac{t}{70} = \frac{1}{2}$$

t = 35 hours

Hence It will take 35 hours in filling the half empty cistern completely.

35. A tap can fill a tank completely in 1 hour 30 min but due to leakage in the tank it takes 2 hours and 15 minutes. What time will be taken to empty the tank due to leakage?

(a) 45 minute (b) 4 hours 30 minute
(c) 1 hours 30 minute (d) 2 hours 45 minute

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (b) 1 hour 30 minute = 90 minute
2 hour 15 minute = 135 minute

According to the question-

emptied portion of cistern in 1 minute because of leakage

$$= \frac{1}{90} - \frac{1}{135} = \frac{3-2}{270} = \frac{1}{270}$$

Hence time taken in emptying the cistern = 270 minute
= 4 hours 30 minute

36. A tap can fill a tank in 25 minutes and the second tap can empty the same tank in 50 minutes. If the both taps are opened together, then tank will be filled ___ minutes?

(a) 55 (b) 45
(c) 20 (d) 50

Ans. (d) : A tap fills in 1 minute $= \frac{1}{25}$ part

second tap fills in 1 minute $= \frac{1}{50}$ part

$$= \frac{1}{25} - \frac{1}{50} = \frac{2-1}{50} = \frac{1}{50}$$

total time taken in filling the tank = 50 minutes

37. A tank has two pipes. Pipe M is to fill the tank and pipe N is to empty the tank. If pipe M takes 45 hours to fill the tank completely and N takes 90 hours to empty the tank completely. Then how much time will it take to fill the half empty tank?

(a) 45 hours (b) 35 hours
(c) 60 hours (d) 40 hours

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (a)
Filled portion of cistern by pipe M and N in 1 hour

$$=\frac{1}{45}-\frac{1}{90}=\frac{2-1}{90}=\frac{1}{90}$$

Time taken by M and N in filling whole cistern = 90 hours

$\therefore$ Time taken in filling half portion of cistern $=\frac{90}{2}=$ 45 hours

38. A tank normally gets filled in 8 hours, but due to leakage in its surface, it takes two hour more to get filled. If the tank is full, then the leakage will empty it in ____?

(a) 20 hours (b) 40 hours
(c) 30 hours (d) 10 hours

RRB Group-D – 12/10/2018 (Shift-II)

Ans : (b) Time taken in filling full of tank = 8 hours

filled portion in 1 hour = $\frac{1}{8}$ parts

Due to leakage, time taken in filling tank = 8 + 2 = 10 hours

filled portion in 1 hour = $\frac{1}{10}$

Emptied portion of tank in 1 hour by leakage

$$=\frac{1}{8}-\frac{1}{10}=\frac{5-4}{40}=\frac{1}{40}\text{ part}$$

Hence due to leakage tank will be empty in 40 hours.

39. A tube can fill a tank in 15 hours. Due to leakage in its bottom, it gets filled in 20 hours. If the tank is fully filled, then due to leakage it will get empty in how much time?

(a) 20 hours (b) 60 hours
(c) 32 hours (d) 40 hours

RRB Group-D – 19/09/2018 (Shift-I)

Ans : (b) Time taken in filling the cistern by the tap = 15 hours

filled portion in 1 hour = $\frac{1}{15}$ part

Because of leakage time taken in filling cistern = 20 hours

Hence filled portion in 1 hour = $\frac{1}{20}$ part

Suppose cistern will be empty in x hours by leakage

so $\frac{1}{x}=\frac{1}{15}-\frac{1}{20}$

$$\frac{1}{x}=\frac{4-3}{60}=\frac{1}{60}$$

x = 60 hours

40. A tank can be filled in 30 minutes. There is a leakage in the tank which can empty the tank in 90 minutes. Therefore the tank will be filled in how much time?

(a) 60 minute (b) 45 minute
(c) 55 minute (d) 50 minute

RRB NTPC 31.03.2016 : 2

Ans : (b) Filled portion of tank in 1 minute

$$=\frac{1}{30}-\frac{1}{90}$$

$$=\frac{3-1}{90}=\frac{1}{45}$$

$\therefore$ Tank will fill in 45 minutes.

41. It takes 12.5 minute for an utensil to get fully filled with juice. By this utensil the children drink juice at a fixed rate due to which the utensil is emptied in 25 minutes. By the present rate how much time will be taken to fill the utensil?

(a) 20 minute (b) 12.5 minute
(c) 25 minute (d) 30 minute

RRB NTPC 18.01.2017 Shift : 3

Ans : (c) Time taken in filling the vessel

$$=\frac{1}{12.5}-\frac{1}{25}=\frac{10}{125}-\frac{1}{25}=\frac{10-5}{125}$$

$$=\frac{5}{125}=\frac{1}{25}$$

hence vessel will fill in 25 minutes.

42. A tap can fill a tank in 20 minutes. If a leakage is capable of emptying the tank in 60 minute, then the tank will be filled in how much time?

(a) 1 hour (b) 45 minute
(c) 30 minute (d) 50 minute

RRB NTPC 06.04.2016 Shift : 1

Ans : (c)

Filled portion by filler tap in 1 minute = $\frac{1}{20}$

and emptied portion by emptying tap in 1 minute

$$=\frac{1}{60}$$

so filled portion by both tap in 1 minute = $\frac{1}{20}-\frac{1}{60}$

$$=\frac{40}{20\times 60}=\frac{1}{30}$$

hence, time taken in filling the tank completely by both = 30 minutes.

43. A Honda motor can fill a sintex tank in 4 hours. There was a leakage in the tank due to which it took six hour to fill the tank. Suppose that the tank is fully filled, in how much time leakage will empty the tank?

(a) 12 hours (b) 15 hours
(c) 11 hours (d) 10 hours

RRB NTPC 22.04.2016 Shift : 3

Ans : (a) Required time $= \frac{1}{4} - \frac{1}{6}$

$= \frac{3-2}{12} = \frac{1}{12}$

Hence leakage will empty the tank in 12 hours.

44. An oil tank takes 15 minutes to be filled. The oil tank is emptied by an outlet pipe which can empty it in 30 minutes. If this outlet pipe remains open, then how much time will it take to fill the tank completely?

(a) 20 minute (b) 25 minute
(c) 30 minute (d) 40 minute

RRB NTPC 28.04.2016 Shift : 3

Ans : (c) If outlet pipe is open then filled part in 1 minute $= \frac{1}{15} - \frac{1}{30} = \frac{30-15}{15\times 30}$

time taken to fill tank $= \frac{30\times 15}{30-15} = \frac{30\times 15}{15} = 30$ minute

45. A pump can fill a tank in 4 hours, but due to a leak, the tank now gets filled in 5 hours. How long will it take the leakage to empty the tank when it is full?

(a) 20 hours (b) 9 hours
(c) 1 hour (d) 4.5 hours

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (a) Let it will take 't' time in emptying the tank. Then-

$$\frac{1}{4} - \frac{1}{t} = \frac{1}{5}$$
$$\frac{1}{t} = \frac{1}{4} - \frac{1}{5}$$
$$\frac{1}{t} = \frac{5-4}{20}$$
$$\frac{1}{t} = \frac{1}{20}$$

tank will be empty in 20 hours

46. One pipe can fill an empty cistern in 4 hours while another can drain the cistern when full in 10 hours. Both the pipes were turned on when the cistern was half-empty. How long will it take to fill the cistern completely?

(a) 6 hours 40 minutes
(b) 5 hours 30 minutes
(c) 4 hours 20 minutes
(d) 3 hours 20 minutes

RRB ALP & Tec. (13-08-18 Shift-III)

Ans : (d) According to the question-

filled portion by first pipe in 1 hour $= \frac{1}{4}$,

emptied portion in 1 hour by second pipe $= \frac{1}{10}$ (First + Second)

filled portion in 1 hour by first and second pipe

$$= \left(\frac{1}{4} - \frac{1}{10}\right) = \frac{5-2}{20}$$
$$= \frac{3}{20} \text{ part}$$

$\because$ half portion of cistern is filled.

hence remaining portion of cistern $= 1 - \frac{1}{2} = \frac{1}{2}$

$\because$ time taken in filling $\frac{3}{20}$ portion = 1 hour

$\therefore$ time taken in filling $\frac{1}{2}$ portion $= \frac{20}{3} \times \frac{1}{2} = \frac{10}{3}$ hours

= 3 hours 20 minute

Type - 1

47. There are three inlets in a tank when the first two are opened then it takes the same time to fill the tank as the third inlet takes to fill the tank. The second inlet takes 5 hours more as compared to previous one and as compared to third inlet it takes 4 hours less. First inlet alone can fill the tank in how many hours?

(a) 15 hours (b) 6 hours
(c) 12 hours (d) 10 hours

RRB JE - 26/06/2019 (Shift-I)

Ans : (a) Let the time taken by the first inlet to fill the tank is x hours.

the second inlet takes time to fill the tank = (x + 5) hours

The third inlet takes time to fill the tank = (x + 9) hours

According to the question-

$$\frac{1}{x} + \frac{1}{x+5} = \frac{1}{x+9}$$
$$\frac{2x+5}{x^2+5x} = \frac{1}{x+9}$$
$$2x^2 + 5x + 18x + 45 = x^2 + 5x$$
$$x^2 + 18x + 45 = 0$$
$$x^2 + 15x + 3x + 45 = 0$$
$$x(x + 15) + 3(x + 15) = 0$$
$$(x + 15) + (x + 3) = 0$$
$$x + 15 = 0 \text{ or } x + 3 = 0$$
$$x = -15 \quad (\because \text{ Time is not negative})$$
$$\therefore x = 15 \text{ hours}$$

Hence the time taken to fill the cistern by the first inlet = 15 hours

48. A tap can fill tank completely in 6 hours. After filling half of the tank three more such tanks are opened. How much time will it take to fill the tank?

(a) 2 hours 20 minutes
(b) 4 hours 30 minutes
(c) 3 hours 25 minutes
(d) 3 hours 45 minutes

RRB JE - 31/05/2019 (Shift-III)

Ans. (d) Filled portion of tank in 1 hour by first tap = $\frac{1}{6}$

filled portion in 1 hour by 4 taps = $\frac{4}{6}=\frac{2}{3}$

Hence four tap can fill the tank in 3/2 hours or 90 minutes.

time taken in filling half tank = $\frac{90}{2}$ = 45 minutes

time taken by first tap to fill the half tank = 3 hours

So total time taken to fill the tank = 3 hours 45 minute

49. Pipe A and C can fill a vacant tank in 7 hrs and 10.5 hours respectively while pipe B can empty the filled tank in 5.25 hours. If the three pipes are opened together when the tank is empty, how many hours will be taken to fill the 2/3 part of the tank?

(a) 14 (b) 21
(c) 12 (d) 15.75

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans : (a) Suppose three pipes will fill the 2/3 portion of cistern in x hours.

$$\Rightarrow \frac{x}{7}+\frac{x}{10.5}-\frac{x}{5.25}=\frac{2}{3}$$

(because it is to fill only 2/3 portion of cistern)

$$\Rightarrow \frac{x}{7}+\frac{10x}{105}-\frac{100x}{525}=\frac{2}{3}$$

$$\Rightarrow \frac{75x+50x-100x}{525}=\frac{2}{3}$$

$$\Rightarrow \frac{125x-100x}{525}=\frac{2}{3}$$

$$\Rightarrow \frac{25x}{525}=\frac{2}{3}$$

$$\Rightarrow \frac{x}{21}=\frac{2}{3}$$

$$\Rightarrow x=\frac{21\times 2}{3}=14$$

$\Rightarrow$ x = 14 hours

50. Two pipes X and Y can fill a gas tank in 60 and 75 minutes respectively. There is an outlet Z. If all the three pipes are used together, then the tank gets full in 50 minutes. How much time will be taken by Z to empty the tank completely?

(a) 100 minutes (b) 75 minutes
(c) 90 minutes (d) 50 minutes

RRB RPF SI – 11/01/2019 (Shift-I)

Ans : (a) Suppose time taken by 'Z' is x minute

filled portion of cistern by 'X' in 1 minute = $\frac{1}{60}$

filled portion of cistern by 'Y' in 1 minute = $\frac{1}{75}$

emptied portion of cistern by 'Z' in 1 minute = $\frac{1}{x}$

As per the question-

$$\frac{1}{60}+\frac{1}{75}-\frac{1}{x}=\frac{1}{50}$$

$$\Rightarrow -\frac{1}{x}=\frac{1}{50}-\left(\frac{1}{60}+\frac{1}{75}\right)$$

$$\Rightarrow -\frac{1}{x}=\frac{1}{50}-\left(\frac{5+4}{300}\right)$$

$$\Rightarrow -\frac{1}{x}=\frac{1}{50}-\frac{9}{300}$$

$$\Rightarrow -\frac{1}{x}=\frac{6-9}{300}=-\frac{3}{300}$$

$$\Rightarrow -\frac{1}{x}=-\frac{1}{100}$$

$\Rightarrow$ x = 100 minute

Hence time taken by 'Z' = 100 minute

51. Pipe A and B can fill a tank in 7 and 10 hours respectively while pipe C can empty the tank in 14 hrs. If all the pipes start together, then how much time will it take to fill the tank?

(a) $6\frac{1}{6}$ hours (b) $6\frac{2}{3}$ hours
(c) $5\frac{1}{2}$ hours (d) $5\frac{5}{6}$ hours

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans. (d) : filled portion of tank by pipe A in 1 hour = $\frac{1}{7}$

filled portion of tank by pipe B in 1 hour = $\frac{1}{10}$

filled portion of tank by pipe C in 1 hour = $\frac{1}{14}$

from the question-

If all the three pipes are opened, then filled portion of tank in 1 hour $=\frac{1}{7}+\frac{1}{10}-\frac{1}{14}$

$$=\frac{10+7-5}{70}=\frac{12}{70}=5\frac{5}{6}$$

Hence all three pipes will fill the tank in $5\frac{5}{6}$ hours

52. Two pipe working individually can fill a tank in 6.3 and 8.4 hours respectively, while the third pipe can empty filled tank in 4.8 hours. If all the 3 pipes are opened together when the tank is empty, then how much time will be taken to fill the tank completely?

(a) 13 hours 20 minutes
(b) 12 hours 18 minutes
(c) 14 hours 18 minutes
(d) 14 hours 24 minutes

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (d) : Filled portion in 1 hour by three pipes

$$= \frac{1}{6.3}+\frac{1}{8.4}-\frac{1}{4.8}$$
$$=\frac{10}{63}+\frac{10}{84}-\frac{10}{48}$$
$$=\frac{10(16+12-21)}{1008}$$
$$=\frac{10\times 7}{1008}=\frac{5}{72}$$

Hence time taken in filling the tank $=\frac{72}{5}=14\frac{2}{5}$ hours

=14 hours 24 minute

53. A, B and C are three valves connected to a tank. A, B together can fill a tank in 6 hours. B and C together can fill the tank in 10 hours. A and C together fill the tank in $7\frac{1}{2}$ hours. A alone will take how much time to fill the tank?

(a) 10 hours (b) 12 hours
(c) 11 hours (d) 13 hours

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (a) filled portion by A + B in 1 hour $=\frac{1}{6}$

filled portion by B + C in 1 hour $=\frac{1}{10}$

filled portion by C + A in 1 hour $=\frac{2}{15}$

Hence by three
filled portion by (A + B + B + C + C+ A) in 1 hour
$=\frac{1}{6}+\frac{1}{10}+\frac{2}{15}$

filled portion by 2(A + B + C) in 1 hour $=\frac{12}{30}$

filled portion by (A + B + C) in 1 hour $=\frac{12}{60}=\frac{1}{5}$

filled portion by A in 1 hour $=\frac{1}{5}-\frac{1}{10}$

filled portion by A in 1 hour $=\frac{2-1}{10}=\frac{1}{10}$

Hence cistern will fill by A in 10 hours.

54. There are three pipes connected to a tank. First pipe can fill the tank in 30 minutes and the second pipe can fill the tank in 45 minutes while the third pipe is to empty the tank. If it takes 27 minutes to fill the tank when all three pipes are opened. In what time the third pipe will empty the tank?

(a) 54 minutes (b) 52 minutes
(c) 50 minutes (d) 56 minutes

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (a) Time taken to fill the tank if three pipes are opened = 27 minute

suppose third pipe empty the tank in x minute then

$$\frac{1}{30}+\frac{1}{45}-\frac{1}{x}=\frac{1}{27}$$
$$\frac{1}{x}=\frac{1}{30}+\frac{1}{45}-\frac{1}{27}$$
$$=\frac{9+6-10}{270}$$
$$=\frac{15-10}{270}$$
$$=\frac{5}{270}$$
$$\frac{1}{x}=\frac{1}{54}$$
$$\boxed{x=54}$$

Hence third pipe will empty the tank in 54 minutes.

55. Tap A and Tap B can fill a tank in 2 and 8 hours respectively. Tap C can empty the tank in 4 hours. If all the taps are opened at a time, then how much time will be taken to fill the tank?

(a) $\frac{8}{3}$ hours (b) $\frac{9}{2}$ hours
(c) 3 hours (d) $\frac{3}{8}$ hours

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (a) filled portion of cistern by tap A in 1 hour = $\frac{1}{2}$

filled portion of cistern by tap B in 1 hour = $\frac{1}{8}$

emptied portion of cistern by tap C in 1 hour $=\frac{1}{4}$

As per the question,

∴ (A+B+C) will fill the tank in an hour $=\frac{1}{2}+\frac{1}{8}-\frac{1}{4}=\frac{3}{8}$ part

∴ Time taken by three taps to fill the cistern = $\frac{8}{3}$ hours

56. There is two entrance in a tank, which can fill the tank in 6 hrs and 8 hrs respectively. By an exit gate the whole tank can be emptied in 10 hours. If all the three pipes are opened together in an empty tank, then how much time will be taken to fill the tank completely?

(a) $5\frac{5}{23}$ hours (b) $6\frac{5}{23}$ hours

(c) $6\frac{5}{46}$ hours (d) $5\frac{5}{46}$ hours

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (a) Filled portion by A+B+C in 1 hour

$$= \frac{1}{6}+\frac{1}{8}-\frac{1}{10}=\frac{23}{120}$$

$\therefore$ time taken in filling whole cistern

$$= \frac{120}{23}=5\frac{5}{23}\text{ hour}$$

57. Two inlet pipes can completely fill a tank in 5 and 7 hours and an outlet can empty the filled tank in 14 hours. If all the three pipes are opened together in an empty tank, then how much time will it take to fill the tank completely?

(a) $4\frac{11}{13}$ hours (b) $5\frac{11}{13}$ hours

(c) $3\frac{13}{19}$ hours (d) $5\frac{2}{13}$ hours

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (c) Filled portion of tank by A in 1 hour $=\frac{1}{5}$

filled portion of tank by B in 1 hour $=\frac{1}{7}$

emptied portion of tank by C in 1 hour $=\frac{1}{14}$

filled portion by three taps in 1 hour $=\frac{1}{5}+\frac{1}{7}-\frac{1}{14}$

$$=\frac{14+10-5}{70}=\frac{19}{70}$$

Hence, tank can be fill by three taps in $3\frac{13}{19}$ hours.

58. In the three pipes the first two pipes can fill a tank in 10.8 hours and 21.6 hours respectively. While the third pipe can empty the filled tank in 18 hours. If the tank is empty and all the three pipes are opened together, then the tank will be filled in how many hours?

(a) 13.2 (b) 12

(c) 14.4 (d) 15.6

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (b) Filled portion by three pipes in 1 hour

$$=\frac{1}{10.8}+\frac{1}{21.6}-\frac{1}{18}=\frac{10}{108}+\frac{10}{216}-\frac{1}{18}$$

$$=\frac{20+10-12}{216}=\frac{18}{216}=\frac{1}{12}\text{ part}$$

Time taken by three pipes in filling the tank = 12 hour

59. Two pipes working together can fill a tank in 3.9 hours and 5.2 hours respectively, while a third pipe, can empty the full tank in 10.4 hours. When the tank is 1/12 filled, all three pipes are opened together. How much time will be taken to fill the tank completely?

(a) 2 hours 45 minutes (b) 2 hours 10 minutes

(c) 2 hours 11 minutes (d) 2 hours 36 minutes

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (d) Remaining part of cistern $=1-\frac{1}{12}=\frac{11}{12}$ part

According to the question-

$$\frac{t}{3.9}+\frac{t}{5.2}-\frac{t}{10.4}=\frac{11}{12}$$

$$\frac{10t}{39}+\frac{10t}{52}-\frac{5t}{52}=\frac{11}{12}$$

$$\frac{40t+30t-15t}{156}=\frac{11}{12}$$

$$\frac{55t}{156}=\frac{11}{12}$$

$$t=\frac{13}{5}=\text{ 2 hours 36 minute}$$

60. Pipe A can fill a tank in 8 hrs, pipe B can fill in P hours and Pipe C can fill in 24 hours. If all the pipes are opened then it takes 2.4 hours to fill the tank completely. Then find the value of P ?

(a) 2 hours (b) 1 hours

(c) 3 hours (d) 4 hours

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (d) Filled portion of cistern by pipe A in 1 hour

$=\frac{1}{8}$ part

filled portion of cistern by pipe B in 1 hour $=\frac{1}{P}$ part

filled portion of cistern by pipe C in 1 hour $=\frac{1}{24}$ part

all the three pipes fill the tank in 2.4 hours.

from the question-

filled part by (A + B + C) in 1 hour $=\frac{1}{8}+\frac{1}{P}+\frac{1}{24}=\frac{1}{2.4}$

$$\frac{1}{P}=\frac{10}{24}-\frac{1}{8}-\frac{1}{24}$$

$$\frac{1}{P}=\frac{10-3-1}{24}$$

$$\frac{1}{P}=\frac{6}{24}$$

$$\boxed{P=4\text{hour}}$$

61. A tank is filled by two taps A and B in 12 hrs and 16 hrs respectively. Complete tank can be emptied by the third tap in 8 hours. If all the taps are opened together at a time then how long will it take to fill the tank?

(a) 24 hours (b) 40 hours
(c) 16 hours (d) 48 hours

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (d) Suppose time taken in filling the empty tank is T

filled portion of tank by A in 1 hour $=\frac{1}{12}$

filled portion of tank by B in 1 hour $=\frac{1}{16}$

emptied portion of tank by third tap in 1 hour $=\frac{1}{8}$

As per the question-

$$\therefore \frac{1}{T}=\frac{1}{12}+\frac{1}{16}-\frac{1}{8}$$

$$\Rightarrow \frac{1}{T}=\frac{4+3-6}{48}$$

$$\Rightarrow \frac{1}{T}=\frac{1}{48}$$

$\Rightarrow$ T = 48 hours

62. Tap A and B fills a tank in 5 hrs and 20 hrs respectively. Tap empties the full tank in 10 hours. If all the three taps are opened together then how long will it take to full the tank?

(a) $\frac{29}{2}$ hours (b) $\frac{17}{5}$ hours
(c) $\frac{20}{6}$ hours (d) $\frac{20}{3}$ hours

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (d) : Tap A fills the tank in 5 hours
Tap B fills the tank in 20 hours
Tap C fills the tank in 10 hours
filled portion of tank by three taps in 1 hour

$$=\frac{1}{5}+\frac{1}{20}-\frac{1}{10}=\frac{4+1-2}{20}=\frac{3}{20}$$

$\because$ total time taken in filling the tank $=\frac{20}{3}$ hours

63. A tap can fill a reservoir in 4 hrs. After the reservoir is half filled 3 more taps of the same size are opened. How much time will be taken by all the taps to fill the tank completely?

(a) 2 hours 30 minutes (b) 2 hours
(c) 1 hours 30 minutes (d) 3 hours

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (a) Suppose reservoir is filled in x hour.
As per the question,

$$\frac{1}{2}+\frac{x}{4}+\frac{x}{4}+\frac{x}{4}+\frac{x}{4}=1$$

$$\frac{4x}{4}=\frac{1}{2} \Rightarrow x=\frac{1}{2} \text{ hour}$$

total time taken in filling reservoir $= 2+\frac{1}{2}$

= 2 hour 30 minute

64. A and C pipes can fill an empty tank in 60 hrs and 48 hrs, respectively while pipe B can empty the full tank in 24 hours. When the tank is full and all the three pipes are opened together then how much time will be taken to fill 1/3 part of the tank?

(a) 240 (b) 160
(c) 80 (d) 120

RRB Group-D – 01/11/2018 (Shift-II)

Ans : (b) Filled portion of tank by pipe A

$=\frac{1}{60}$ part

filled portion of tank by pipe C in 1 hour $=\frac{1}{48}$ part

emptied portion of tank by pipe B in 1 hour $=\frac{1}{24}$ part

So emptied portion by three pipe in 1 hour

$$=\frac{1}{24}-\left(\frac{1}{60}+\frac{1}{48}\right)$$

$$=\frac{10-(4+5)}{240}=\frac{1}{240}\text{ part}$$

So time taken in emptying the $\frac{2}{3}$ portion of tank = time taken to fill the $\frac{1}{3}$ portion of tank $=\frac{2/3}{1/240}=160$ hours

65. A, B and C three tubes are attached with a reservoir. A and B together can fill the reservoir in 6 hours. B and C together can fill the reservoir in 10 hours. A and C together can fill the reservoir in $7\frac{1}{2}$ hours. How much time will be taken by C alone to fill the tank?

(a) 25 hours (b) 15 hours
(c) 40 hours (d) 30 hours

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (d)
Work done in 1 hour by A + B

$$=\left(\frac{1}{A}+\frac{1}{B}\right)=\frac{1}{6}\text{ part ..... (1)}$$

work done in 1 hour by B + C

$$=\left(\frac{1}{B}+\frac{1}{C}\right)=\frac{1}{10}\text{ part ..... (2)}$$

work done in 1 hour by A + C

$$=\left(\frac{1}{A}+\frac{1}{C}\right)=\frac{1}{\frac{15}{2}}=\frac{2}{15}\text{ part..... (3)}$$

on adding of equation (1), (2), (3)

$$\left(\frac{1}{A}+\frac{1}{B}\right)+\left(\frac{1}{B}+\frac{1}{C}\right)+\left(\frac{1}{A}+\frac{1}{C}\right)=\frac{1}{6}+\frac{1}{10}+\frac{2}{15}$$

$$2\left(\frac{1}{A}+\frac{1}{B}+\frac{1}{C}\right)=\frac{5+3+4}{30}$$

$$2\left(\frac{1}{A}+\frac{1}{B}+\frac{1}{C}\right)=\frac{12}{30}$$

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{12}{30\times 2}=\frac{1}{5}\text{.....(4)}$$

on subtracting the equation (1) from equation (4)

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}-\left(\frac{1}{A}+\frac{1}{B}\right)=\frac{1}{5}-\frac{1}{6}$$

$$\frac{1}{C}=\frac{6-5}{30}=\frac{1}{30}$$

$$\frac{1}{C}=\frac{1}{30}$$

time taken in filling the reservoir by C = 30 hours

66. Two pipes can fill a tank in 5 hrs and 3 hrs respectively while the third pipe can empty the tank in 7.5 hours. When 1/10 part of the tank was filled then all the three pipes were opened together, then how long will it take to fill the tank?

(a) 2 hours 20 minutes (b) 2 hours
(c) 2 hours 15 minutes (d) 2 hours 30 minutes

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (c) :

Emptied portion of cistern $= 1-\frac{1}{10}=\frac{9}{10}$ part

filled portion by three pipes in 1 hour

$$=\frac{1}{5}+\frac{1}{3}-\frac{1}{7.5}$$

$$=\frac{1}{5}+\frac{1}{3}-\frac{10}{75}$$

$$=\frac{15+25-10}{75}=\frac{30}{75}=\frac{2}{5}\text{ part}$$

$\because$ filled portion by three pipes in 1 hour $=\frac{2}{5}$

time taken in filling $\frac{2}{5}$ portion = 1 hour

time taken in filling $\frac{9}{10}$ portion $=\frac{5}{2}\times\frac{9}{10}$

= 2 hour 15 minute

67. Pipes A, B and C are attached to an empty cistern. While the first two can fill the cistern in 4 and 10 hours, respectively, the third can drain the cistern, when filled, in 6 hours. If all the three pipes are opened simultaneously when the cistern is half-full, how many hours will be needed to fill the cistern?

(a) $\frac{30}{11}$ (b) $\frac{60}{11}$
(c) $\frac{120}{11}$ (d) $\frac{90}{11}$

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (a) According to the question,

$$=\frac{1}{4}+\frac{1}{10}-\frac{1}{6}$$

$$=\frac{15+6-10}{60}=\frac{11}{60}\text{ part}$$

time taken in filling whole cistern $=\frac{60}{11}$ hour

When cistern is half fill then time taken

$$=\frac{60}{2\times 11}=\frac{30}{11}\text{ hour}$$

68. Out of three pipes the first two can fill an empty tank in 9 and 18 hours, respectively while the third can drain the filled tank in 15 hours. If all the three pipes are opened when the tank is empty, after how many hours will the tank be fill?

(a) 10 (b) 12
(c) 11 (d) 13

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (a)

Filled portion of cistern by pipe A in 1 hour $=\frac{1}{9}$

filled portion of cistern by pipe B in 1 hour $=\frac{1}{18}$

and emptied portion of cistern by pipe C in 1 hour $=\frac{1}{15}$

$\therefore$ filled portion of cistern by (A+B+C) in 1 hour

$$=\frac{1}{9}+\frac{1}{18}-\frac{1}{15}$$

$$=\frac{10+5-6}{90}$$

$$=\frac{9}{90}\Rightarrow\frac{1}{10}$$

so time taken in filling whole cistern= 10 hour

69. Raghu's tanker can fill a cistern in 4 hours. After half the cistern is filled, three more similar tankers are opened. what is the total time taken to fill the cistern completely ?

(a) 2 h 40 min (b) 2 h
(c) 3 h (d) 2 h 30 min

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (d)

$\because$ Raghu's tanker fills half portion of reservoir in 2 hours.

$\therefore$ Three more tanker just like that = total 4 tanker will fill $\frac{1}{2}$ portion of cistern $=\frac{4}{4}\times\frac{1}{2}=30$ minutes

hence time taken in filling whole cistern = 2 hours 30 minute

70. Pipes A and C can fill an empty cistern in 32 and 48 hours, respectively while Pipe B can drain the filled cistern in 24 hours. If the three pipes are turned on together when the cistern is empty, how many hours will it take for the cistern to be $\frac{2}{3}$ full?

(a) 96 (b) 64
(c) 72 (d) 48

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (b) Filled portion by pipe 'A' in 1 hour $=\frac{1}{32}$

filled portion by pipe 'C' in 1 hour $=\frac{1}{48}$

emptied portion by pipe 'B' in 1 hour $=\frac{1}{24}$

filled portion by three pipes in 1 hour $=\frac{1}{32}+\frac{1}{48}-\frac{1}{24}$

$=\frac{3+2-4}{96}=\frac{1}{96}$

$\because$ It takes 1 hour in filling $\frac{1}{96}$ portion

$\therefore$ time taken in filling $\frac{2}{3}$ portion $=96\times\frac{2}{3}=64$ hours

71. Pipes A, B and C are attached to an empty cistern. While the first two can fill the cistern in 4 and 10 hours, respectively, the third can drain the cistern, when filled, in 6 hours. If all the three pipes are opened simultaneously when the cistern is three-fifth full, how many hours will be needed to fill the cistern?

(a) $\frac{36}{11}$ (b) $\frac{48}{11}$
(c) $\frac{60}{11}$ (d) $\frac{24}{11}$

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (d) A, B and C will fill the cistern in 1 hour

$=\frac{1}{4}+\frac{1}{10}-\frac{1}{6}=\frac{15+6-10}{60}=\frac{11}{60}$ part

$\therefore$ All three pipes will fill the cistern completely

$=\frac{60}{11}$ hour

Remaining empty portion of cistern $=1-\frac{3}{5}=\frac{2}{5}$ part

$\therefore$ Time taken in filling remaining empty portion of cistern $=\frac{60}{11}\times\frac{2}{5}=\frac{24}{11}$ hour

Type - 1

72. Pipes P and Q can fill a tank in 6 hrs and 4 hrs respectively. Pipe P is opened at 7 O'clock and P and Q is opened one by one for an hour. Tank will be filled in how many hours?

(a) 6 hours
(b) 3 hours
(c) 4 hours 30 minutes
(d) 5 hours

RRB JE - 29/05/2019 (Shift-II)

Ans : (d) L.C.M of 6 and 4 = 12 unit work
1 hour work of P = 2 unit
1 hour work of Q = 3 unit
2 hour work of (P + Q) = 5 unit
Hence 2 × 2 hours work = 5 × 2 = 10 unit
Remaining work = 2 unit
$\therefore$ the time taken by P to finish 2 unit work = 1 hour
$\therefore$ the time taken to fill the tank = 2 × 2 + 1 = 5 hours

73. A tank can be filled by a tap in 10 hours, although due to a leakage it takes 11 hours to fill the tank. The tap is closed when the tank is full. In how much time the tank will be emptied due to leakage?

(a) 130 (b) 110
(c) 100 (d) 50

RRB RPF Constable – 20/01/2019 (Shift-I)

Ans. (b) : Filled part in 1 hour by tap = $\frac{1}{10}$ part

filled part in 1 hour = $\frac{1}{11}$ part

It takes x hours to empty the tank due to leakage.
According to the question,

$\frac{1}{10}-\frac{1}{x}=\frac{1}{11}$

$\Rightarrow\frac{1}{x}=\frac{1}{10}-\frac{1}{11}$

$=\frac{11-10}{110}$

$\Rightarrow\frac{1}{x}=\frac{1}{110}$

x = 110 hours

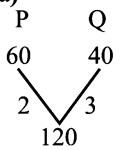
Hence the time taken for the tank to be emptied = 110 hours

74. Two taps P and Q individually take 60 min and 40 min to fill a tank. If tap Q is opened for the first half an hour and for remaining time tap P is opened, then how long will it take to fill the tank?

(a) 45 minutes (b) 32 minutes
(c) 36 minutes (d) 48 minutes

RRB JE - 30/05/2019 (Shift-I)

Ans : (a)



Let the total capacity of the tank is 120 liters.
[L.C.M of (60, 40) = 120]

∵ Q fills 3 liters of water in 1 minute.

∴ Q will fill in 30 minutes = 3 × 30 = 90 *l*

Remaining amount of the tank = 120 – 90 = 30 *l*

∵ P fills the remaining volume of the tank

$$\Rightarrow \frac{30}{2} = 15 \text{ min.}$$

required time = 30 + 15 = 45 min.

75. Two pipes P and Q can fill a tank in 32 min and 48 min. Both pipes are opened and after some time pipe Q is closed, then tank gets filled in 24 minutes. When was pipe Q closed?

(a) 12 minutes (b) 15 minutes
(c) 10 minutes (d) 16 minutes

RRB JE - 01/06/2019 (Shift-III)

Ans. (a) Filled part in 1 minute by pipe P $= \frac{1}{32}$ part

filled part in 1 minute by pipe Q $= \frac{1}{48}$ part

Pipe P will work from the starting to end.

filled part by P in 24 min. due to leakage $= \frac{24}{32} = \frac{3}{4}$ part

remaining part $= 1 - \frac{3}{4} = \frac{1}{4}$ part

time taken to fill remaining part by Q

$$= \frac{1}{4} \times 48 = 12 \text{ min.}$$

Hence pipe Q was closed after 12 min.

76. Tap P and Q can fill a bucket individually in 12 min and 15 min. Both the taps are opened, but after 3 minutes tap P is closed. What extra time will be taken by Q to fill the bucket?

(a) 6 minutes 15 sec.
(b) 8 minutes 15 sec.
(c) 6 minutes 30 sec.
(d) 9 minutes

RRB JE - 27/06/2019 (Shift-III)

Ans : (b) Let the extra time taken for tap Q to fill the bucket is t minute.
According to the question,

$$\frac{3}{12} + \frac{3+t}{15} = 1$$

$$\frac{3+t}{15} = 1 - \frac{1}{4}$$

$$\frac{3+t}{15} = \frac{3}{4}$$

12 + 4t = 45

4t = 33

$$t = \frac{33}{4}$$

$$t = 8\frac{1}{4} = 8 \text{ min } 15 \text{ sec.}$$

77. Two pipes P and Q fill a tank in 15 min and 20 min respectively. Both are opened but after 4 minute P is closed. In how much time tank is filled from the beginning?

(a) 16 minutes
(b) 16 minutes 20 sec.
(c) 14 minutes 20 sec.
(d) 14 minutes 40 sec.

RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (d) Let extra time taken by Q to fill the tank = t min.

from question,

$$\frac{4}{15} + \frac{t+4}{20} = 1$$

$$\frac{16+3t+12}{60} = 1$$

3t + 28 = 60

3t = 32

t = 10 min. 40 sec.

total time taken = 10 min 40 sec. + 4 min.
= 14 min. 40 sec.

78. Two pipes A and B can fill a tank in 12 min & 16 min respectively. Both pipes are opened together, but pipe A closed 4 minutes before the tank is filled. In how many minutes the tank is completely filled?

(a) 9 minutes 8 sec
(b) 10 minutes 9 sec
(c) 11 minutes 9 sec
(d) 11 minutes 29 sec

RRB NTPC 29.04.2016 Shift : 2

Ans : (a) Let it takes T minute to fill the tank.
According to the question,

$$\frac{T-4}{12} + \frac{T}{16} = 1$$

$$\Rightarrow \frac{4(T-4)+3T}{48} = 1$$

$\Rightarrow$ 4T – 16 +3T = 48

$\Rightarrow$ 7T – 16 = 48

$\Rightarrow$ 7T = 48 + 16

$\Rightarrow$ 7T = 64

$$\Rightarrow T = \frac{64}{7} = 9\frac{1}{7} \text{ minute} = 9 \text{ min. } \frac{1}{7} \times 60 \text{ sec.}$$

= 9 min 8.5 sec ≈ 9 min 8 sec.

79. Pipes A and B work together then they can fill an empty tank in 24 hours. If working together for 8 hours B is closed but A remains open, then it will take 28 hours to fill the tank. A alone will fill the tank in how much time?

(a) 30 hours (b) 31 hours
(c) 28 hours (d) 29 hours

RRB RPF SI – 12/01/2019 (Shift-III)

Ans. (a) : Let A will take t hours to fill the tank alone.
According to the question,

$$\frac{8}{24}+\frac{20}{t}=1$$

$$\frac{20}{t}=1-\frac{1}{3}$$

$$\frac{20}{t}=\frac{2}{3}$$

t = 30 hours

Hence A alone will take 30 hours to fill the tank.

80. Two pipes X and Y can fill a tank in 24 hrs and 32 hrs. If both the pipes are opened together then at what time first pipe should be closed that it may take only 16 hours to fill the tank?

(a) After 18 hours (b) After 10 hours
(c) After 15 hours (d) After 12 hours

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (d) : Let the first pipe should be closed after x hours.
According to the question,

$$\Rightarrow \quad \frac{x}{24}+\frac{16}{32}=1$$

$$\Rightarrow \quad \frac{x}{24}+\frac{1}{2}=1$$

$$\Rightarrow \quad \frac{x}{24}=\frac{1}{2}$$

x = 12 hours

Hence first pipe should be closed after 12 hours.

81. Two gas filling tube A and B can fill a cylinder in 12 and 15 minutes respectively. But a third tube can empty it in 6 minutes. First two tubes are opened for 5 minutes and then the third tube is also opened. In what time cylinder will be empty?

(a) 70 minutes (b) 45 minutes
(c) 60 minutes (d) 30 minutes

RRB Group-D – 12/10/2018 (Shift-II)

Ans : (b) Tube A → 12 minutes
tube B → 15 minutes [L.C.M. of 12 and 15 = 60 unit]
tube C → in 6 minutes

tube A → 12 min → 60 unit
A → 1 min → 5 unit
tube B → 15 min → 60 unit
B → 1 min → 4 unit

filled part by both tubes (A + B) in 1 minute = 9 unit
both tubes (A + B) are opened for first 5 minutes, then work done = 5 × 9 = 45 unit

C will empty in 1 minute $= \frac{60}{6}$ = 10 unit

when the C tube is also opened.
the work of (A + B – C) tube in 1 minute = 5 + 4 – 10
= –1 unit
therefore it will take 45 minutes to empty 45 units.

82. Two inlet pipes A and B can fill an empty tank in 22 and 33 hours respectively. They were opened together but 3 hours before when tank was completely filled, pipe A was stopped. How much time will be taken by both the pipes to fill the tank?

(a) 16.2 (b) 15
(c) 14.2 (d) 16

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) Let the time taken by both pipes to fill the tank in x hours.
from question,

$$\frac{x-3}{22}+\frac{x}{33}=1$$

$$\frac{3x-9+2x}{66}=1$$

5x – 9 = 66
5x = 75
x = 15
therefore it takes 15 hours to fill the tank by both pipes.

83. Two pipes A and B can fill an empty tank in 1.8 hrs and 2.7 hrs respectively. Without functioning of any pipe, Pipe C can empty the whole tank in 4.5 hrs. In the beginning pipe A and pipe C are opened after the tank is emptied. After some hours pipe A is closed and pipe B is opened. In this way total 5.5 hours is taken to fill the tank. For how much time pipe B remained open.

(a) 2.7 (b) 5
(c) 3 (d) 4.5

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (d) Filled part by pipe A in 1 hour $= \frac{1}{1.8}$

filled part by pipe B in 1 hour $= \frac{1}{2.7}$

emptied part by pipe C in 1 hour $= \frac{1}{4.5}$

Let pipe A is opened for x hours.
so pipe B is opened for (5.5–x) hours

$$\frac{x}{1.8}+\frac{(5.5-x)}{2.7}-\frac{5.5}{4.5}=1$$

$$=\frac{15x+55-10x-33}{27}=1$$

5x = 27 + 33 – 55
5x = 5
x = 1
Hence pipe A is opened for 1 hour
so pipe B worked = (5.5–1) = till 4.5 hours

84. Three flood gate A, B and C can fill a reservoir in 6 hours. After working for two hours together C was closed, remaining part is filled by flood gates A and B in 7 hours. What time will be taken by flood gate C to fill the reservoir?

(a) 16 (b) 12
(c) 14 (d) 10

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (c) Filled part by all the three flood gates A, B, C in 1 hour = $\frac{1}{6}$ part

$\therefore$ filled part by all three flood gates in 2 hour = $\frac{1}{6}\times 2=\frac{1}{3}$ part

$\therefore$ remaining part $=1-\frac{1}{3}=\frac{2}{3}$ part

filled part by A and B in 7 hours $=\frac{2}{3}$

filled part by A and B in 1 hour $=\frac{2}{21}$ part

part filled by C in 1 hour $=\frac{1}{6}-\frac{2}{21}$

$=\frac{7-4}{42}=\frac{3}{42}=\frac{1}{14}$

time taken by C to fill the reservoir = 14 hours

85. There are two entrance pipes in a tank by which the tank can be filled in 4 hours and 6 hours respectively. By an exit pipe the whole tank can be emptied in 8 hours. Both the entrance pipes are opened for an hour and then closed. Then after all the three pipes are opened together. Remaining part of the tank can be filled in ___ hours.

(a) 3 hours (b) $3\frac{1}{4}$ hours
(c) 2 hours (d) $5\frac{1}{3}$ hours

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c) Let the time taken to fill the remaining part of the tank = t hours

tank filled in 1 hours by both entrance pipes $=\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$ part

remaining part $=1-\frac{5}{12}=\frac{7}{12}$ part

According to the question,

$$\frac{t}{4}+\frac{t}{6}-\frac{t}{8}=\frac{7}{12}$$

$$\frac{6t+4t-3t}{24}=\frac{7}{12}$$

$$\frac{7t}{24}=\frac{7}{12}$$

$$t=\frac{7\times 24}{7\times 12}$$

t = 2 hours

Hence the remaining part of the tank will be filled in 2 hours.

86. Three pipes A, B and C can fill a tank in 4 hours. Pipe A and pipe B can together fill the tank in 9 hours. If the three pipes A, B and C are opened together and after 3 hours pipe A and B are closed, then how much time will be taken by pipe C alone to fill the tank?

(a) 2 (b) 1.8
(c) 2.25 (d) 1.5

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (b)

Filled part by (A+B+C) in 1 hour $=\frac{1}{4}$

filled part by (A+B) in 1 hour $=\frac{1}{9}$

filled part by C in 1 hour $=\frac{1}{4}-\frac{1}{9}=\frac{5}{36}$

time taken to fill whole tank by pipe C $=\frac{36}{5}$ hour

let C does the remaining work in t time.

According to the question,

$$3\left(\frac{1}{A}+\frac{1}{B}+\frac{1}{C}\right)+\frac{t}{C}=1$$

$$\frac{t}{C}=1-\frac{3}{4}$$

$$t=\frac{36}{5}\times\frac{1}{4}=\frac{9}{5}$$

t = 1.8

the time taken by C to fill the remaining part is 1.8 hours

87. Inlet pipe P can fill any tank in 7 hrs while an outlet pipe Q can empty a whole tank in 6 hrs. If only P is opened for first 3 hrs and then Q is also opened, then how many hours is needed to empty the tank?

(a) 24 (b) 28
(c) 21 (d) 18

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (d) Time taken by P and Q to fill the tank $=\frac{7\times 6}{7-6}=42$ hours

filled part of the tank by P in 1 hour $=\frac{1}{7}$ part

filled part of the tank by p in 3 hours $=\frac{3}{7}$ part

time taken to empty $\frac{3}{7}$ part by tank Q $=\frac{3}{7}\times 42$

= 18 hours

88. Two gas filling tube A and B can fill a gas cylinder in 15 min and 40 min respectively. Both the tubes are opened together but after 4 minutes tube A is closed. What will be the total time to fill the cylinder?

(a) 14 minutes 40 sec.
(b) 10 minutes 10 sec.

(c) 20 minutes 10 sec.
(d) 29 minutes 20 sec.

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (d) : Let the cylinder will be completely fill in time t.
According to the question,

$$\frac{4}{15}+\frac{t}{40}=1, \qquad \frac{32+3t}{120}=1$$

3t = 120–32

$$t=\frac{88}{3}$$

t = 29 min 20 sec.

89. Pipe A can fill an empty reservoir in 6.8 hrs while pipe B can fill it in 10.2 hrs. Only pipe B is opened for 1.7 hours, after which pipe A is also opened. What is the total time to fill the reservoir?

(a) 5 hours 6 minutes (b) 5 hours 10 minutes
(c) 5 hours 12 minutes (d) 5 hours 5 minutes

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (a) : Filled part by A in 1 hour = $\frac{1}{6.8}$

filled part by B in 1 hour = $\frac{1}{10.2}$

filled part by B in 1.7 hours = $\frac{1.7}{10.2}=\frac{1}{6}$

Remaining part = $1-\frac{1}{6}=\frac{5}{6}$ part

filled part by (A+B) in 1 hour

$$=\left(\frac{1}{6.8}+\frac{1}{10.2}\right)=\frac{25}{102}\text{ part}$$

time taken by (A+B) to fill $\frac{5}{6}$ part = $\frac{5}{6}\times\frac{102}{25}$ part

$$=\frac{17}{5}=3.4\text{ hours}$$

total time taken = 1.7+3.4= 5.1 hours
or 5 hour 6 minutes

90. Pipe C and D alone can fill a tank in 4 and 5 hrs respectively. If pipe C, is closed after 3 hours and at the same time pipe D is opened, then the tank will be filled in how many hours?

(a) 1.25 (b) 1
(c) 1.5 (d) 0.8

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (a) Filled part by pipe C in 1 hour = $\frac{1}{4}$

Filled part by pipe C in 3 hours = $3\times\frac{1}{4}=\frac{3}{4}$

Remaining part = $1-\frac{3}{4}=\frac{1}{4}$ part

Time taken by pipe D to fill whole tank = 5 hours

Hence time taken by pipe D to fill remaining $\frac{1}{4}$ part

$$=5\times\frac{1}{4}=\frac{5}{4}=1.25\text{ hours}$$

91. Pipe A alone can fill a tank in 10 hrs. Pipe B can fill the same tank in 12 hours. These pipes are opened together then it is known that due to a leakage in the tank an extra time of $\frac{60}{231}$ hours will be taken to fill the tank completely. If both the pipes are closed, then calculate the time to empty the full tank by the leakage?

(a) 110 hours (b) 132 hours
(c) 143 hours (d) 120 hours

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d) Filled part of the tank by pipe A in 1 hour

$$=\frac{1}{10}$$

Filled part of the tank by pipe B in 1 hour = $\frac{1}{12}$

Let the tank will be empty in x hours due to leakage.

∴ Empty part of tank in 1 hour due to leakage = $\frac{1}{x}$

Filled part of tank by (A + B) in 1 hour

$$=\left(\frac{1}{10}+\frac{1}{12}\right)=\frac{11}{60}$$

So pipes A and B together can fill the tank in $\frac{60}{11}$ hours

Total time taken to fill the tank due to leakage

$$=\frac{60}{11}+\frac{60}{231}=\frac{1320}{231}\text{ hour}$$

from question,

$$\frac{1}{A}+\frac{1}{B}-\frac{1}{x}=\frac{231}{1320}$$

$$\frac{1}{10}+\frac{1}{12}-\frac{1}{x}=\frac{231}{1320}$$

$$\frac{1}{x}=\frac{11}{60}-\frac{231}{1320}$$

$$\frac{1}{x}=\frac{242-231}{1320}=\frac{11}{1320}$$

$$x=\frac{1320}{11}=120\text{ hours}$$

Hence the whole tank should be empty in 120 hours by leakage.

92. Two inlet pipe A and B can fill a reservoir in 2.5 and 15 hrs respectively, while pipe C can empty the full reservoir in 7.5 hours. After the reservoir was empty pipe A, B and C were opened together, but pipe B was closed after sometime, due to which it took 3.5 hours to fill the reservoir. Pipe B was started for how many hours?

(a) 1 hour (b) 2 hours
(c) 1.5 hour (d) 0.5 hour

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (a) Let pipe B was opened for x hours.
According to the question,

$$\frac{1}{A}+\frac{1}{B}-\frac{1}{C}=1$$

$$\frac{3.5}{2.5}+\frac{x}{15}-\frac{3.5}{7.5}=1$$

$$\frac{7}{5}+\frac{x}{15}-\frac{7}{15}=1$$

$$\frac{x}{15}=1-\frac{7}{5}+\frac{7}{15}=\frac{15-21+7}{15}$$

$$\frac{x}{15}=\frac{1}{15} \Rightarrow x = 1 \text{ hour}$$

93. Three pumps can fill a water tank individually in 15, 20 and 30 hrs respectively. All three pumps were opened together at 8 A.M. At that time 1/5 of tank was filled. After 4 hours first pump was closed. After next two hours the third pump was also closed. At which time tank will be fully filled?

(a) 2:30 p.m. (b) 2:40 p.m.
(c) 3:20 p.m. (d) 1:20 a.m.

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (b) : Filled part of tank by first pump in 1 hour $=\frac{1}{15}$ part

Filled part of tank by second pump in 1 hour $=\frac{1}{20}$ part

Filled part of tank by third pump in 1 hour $=\frac{1}{30}$ part

Pre-filled part of the tank $=\frac{1}{5}$ (given)

Remaining part $=1-\frac{1}{5}=\frac{4}{5}$

Let it will take x hours to fill the tank.
According to the question,

$$\frac{4}{15}+\frac{x}{20}+\frac{6}{30}=\frac{4}{5}$$

$$\frac{16+3x+12}{60}=\frac{4}{5}$$

$28 + 3x = 48$
$3x = 20$

$x=\frac{20}{3}$ or 6 hour 40 minutes

Hence required time = 8 am + 6 hour 40 minutes = 2:40 pm

94. Pipe A can fill a tank in 15 hours while pipe B takes 25 hours to fill it. In the beginning only pipe A was started for some time and again it was closed and then just pipe B was started. Overall it takes 19 hours to fill the tank completely. Pipe A was started for how much time?

(a) 9 hours (b) 6 hours
(c) 7.5 hours (d) 3 hours

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (a) Let the pipe A was opened for x hour.
According to the question,

$$\frac{x}{15}+\frac{19-x}{25}=1$$

$$\frac{5x+57-3x}{75}=1$$

$2x + 57 = 75$
$2x = 75 - 57$
$2x = 18$
$x = 9$

Hence pipe A was opened for 9 hours

95. Two inlet pipe and one outlet pipe is present in a tank. The tank can be filled in 6 and 8 hrs respectively by the inlets individually and can be emptied in 10 hrs by the outlet pipe. If the inlet pipe is opened for 1 hour and then closed, and then all the three pipes are opened together, then how much time will be taken to fill the remaining part of the tank?

(a) 75/23 hours (b) 82/23 hours
(c) 85/23 hours (d) 72/23 hours

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (c) Let it will take n hour to fill the remaining tank.
According to the question,

$$\frac{n+1}{8}+\frac{n+1}{6}-\frac{n}{10}=1$$

$$\Rightarrow \frac{6n+6+8n+8}{48}-\frac{n}{10}=1$$

$$\Rightarrow \frac{7n+7}{24}-\frac{n}{10}=1$$

$$\Rightarrow \frac{70n+70-24n}{240}=1$$

$$\Rightarrow \frac{35n+35-12n}{120}=1$$

$$\Rightarrow 23n+35=120$$

$23 n = 85$

$$\boxed{n=\frac{85}{23}\text{hours}}$$

96. Pipe A alone can fill a tank in 4.5 hours, while along with pipe B it can be filled in 2.25 hours. If only pipe A is opened for half an hour after which pipe B is also opened, then how much time will be taken to fill the tank?

(a) 2 hours 30 minutes
(b) 2 hours 15 minutes
(c) 2 hours
(d) 2 hours 20 minutes

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (a) : Let the pipe B will fill the tank in x hours

Filled part of the tank by A in 1 hour $= \frac{1}{4.5}$

Filled part of the tank by A and B in 1 hour $= \frac{1}{2.25}$

$\therefore \quad \frac{1}{4.5} \times \frac{1}{2} + \frac{x}{2.25} = 1$

$\frac{10}{90} + \frac{100x}{225} = 1$

$\frac{1}{9} + \frac{4x}{9} = 1$

$1 + 4x = 9$

$\boxed{x = 2}$ hours

Hence total time to fill the tank =

2 hour + 30 minutes = 2 hour 30 minutes

97. Pipe A can fill an empty cistern in 4 hours while along with Pipe B it can be filled in 3 hours. Only Pipe A is turned on for an hour after which Pipe B is also turned on. How much total time will it take to fill up the cistern?

(a) 3 hours
(b) 3 hours 15 minutes
(c) 3 hours 25 minutes
(d) 3 hours 20 minutes

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (c) : Filled tank by pipe A in 1 hour $= \frac{1}{4}$ part

Filled tank by A and B in 1 hour $= \frac{1}{3}$ part

Emptied part of tank $= 1 - \frac{1}{4} = \frac{3}{4}$ part

$\frac{1}{3}$ part of tank can be filled by (A + B) = in 1 hour.

$\frac{3}{4}$ part of tank can be filled by both = $3 \times \frac{3}{4} = 2.25$

total time = 1 + 2.25 = 3.25 hours

98. Two pipes A and B can fill an empty cistern in 32 and 48 hours, respectively. Pipe C can drain the entire cistern in 64 hours when no other pipe is in operation. Initially, when the cistern was empty, Pipe A and Pipe C were turned on. After a few hours, Pipe A was turned off and Pipe B was turned on instantly. In all it took 112 hours to fill the cistern. For how many hours was Pipe B turned on ?

(a) 72 (b) 70 (c) 77 (d) 84

Ans : (a) Let tap A is closed after x hours and tap B is opened, so it can be assumed that tap A was opened for x hours, tap B was opened for (112-x) hours and C was opened for 112 hours.

then $\boxed{x.A + B(112 - x) - C \times 112 = 1}$

$\Rightarrow \quad \frac{x}{32} + \frac{112 - x}{48} - \frac{112}{64} = 1$

$\Rightarrow \quad \frac{6x + (112 - x)4 - 112 \times 3}{192} = 1$

$\Rightarrow \quad 6x + 448 - 4x - 336 = 192$

$\Rightarrow \quad 2x + 112 = 192$

$2x = 192 - 112$

$2x = 80$

$x = 40$ hours

$\because$ Pipe B was kept running at 112-x hours.

Hence the required time = 112– x = 112 – 40 = 72 hours.

Type - 5

99. Pipe A can empty whole tank in 5.6 hours, while pipe B, can fill the same tank in 7 hours. In the situation of tank being full, if the pipes A and B is opened one by one for an hour, then how much time can be taken to empty the tank?

(a) 48 hours (b) 56 hours
(c) 55 hours (d) 47 hours

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (d)

A = (–)5.6 (–2)
B = (+) 7 (+5.6)
39.2

when, (–) outlet
(+) inlet

work done by (A + B) in 2 hours $= \frac{+5.6}{(-1.4)}$

work done by (A + B) in 46 hours = –1.4 × 23
= 32.2 unit

Remaining work = 39.2 – 32.2 = 7 unit

Now in 47 hours, A will drain 7 unit water.

So total time = 46 + 1 = 47 hours

100. An inlet pipe and outlet pipe are taking turns to fill and drain a cistern, respectively, for an hour each at a time, starting with the inlet pipe when the cistern is empty. It takes the inlet pipe 15 hours to fully fill the empty cistern, whereas the outlet pipe can drain the filled cistern completely in 21 hours. How many hours will it take for the cistern to be full?

(a) 100 (b) 52.5
(c) 105 (d) 99

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (d) : Filled portion of cistern in 2 hours $= \frac{1}{15} - \frac{1}{21}$

$= \frac{1}{15} - \frac{1}{21} = \frac{7-5}{105} = \frac{2}{105}$

On multiplying by 49.

Filled portion of cistern in 98 hours $= \frac{2}{105} \times 49 = \frac{14}{15}$

$\therefore$ Remaining portion $= 1 - \frac{14}{15} = \frac{1}{15}$

Hence A will now fill $\frac{1}{15}$ part in 1 hour.

So, it will take total 99 hours.

101. Pipe A can fill an empty tank in 1.5 hours while pipe B can empty a full tank in 2.4 hours. When the tank is empty then pipe A is opened for 1 hour and then closed. Now B pipe is opened for 1 hour to empty the tank and then closed. The tank was opened and closed alternately for one hour every time. How much time will it take to fill the tank completely?

(a) 4.75 hours (b) 6.375 hours
(c) 7 hours (d) 8 hours

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (d) : work done by pipe A in 1 hour = $\frac{1}{1.5}$ part

work done by pipe B in 1 hour = $\frac{1}{2.4}$ part

filled part in 2 hours

$$= \frac{1}{1.5} - \frac{1}{2.4} = \frac{2.4-1.5}{3.60} = \frac{.9}{3.6} = \frac{9}{36} = \frac{1}{4} \text{ part}$$

so the time taken to fill the entire part = $\frac{4\times 2}{1} = 8$ hours

102. Starting from the inlet pipe in the reservoir, an inlet pipe and an outlet pipe take an hour to fill and empty the reservoir. The inlet pipe takes 11.25 hours to fill the empty reservoir and the outlet pipe can empty the filled reservoir completely in 22.5 hours. How much time will be taken to fill the reservoir?

(a) 44 hours
(b) 45 hours
(c) 43 hours 50 minutes
(d) 42 hours 45 minutes

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (d) Filled part by inlet pipe in 1 hour = $\frac{1}{11.25}$ part

emptied part by outlet pipe in 1 hour = $\frac{1}{22.5}$ part

filled part in 2 hours = $\frac{1}{11.25} - \frac{1}{22.5} = \frac{2-1}{22.5}$

$= \frac{1}{22.5}$

Water filled in 2 × 21 hours = $\frac{1}{22.5} \times 21$

Water filled in 42 hours = $\frac{42}{45}$ part

remaining part of tank = $1 - \frac{42}{45} = \frac{3}{45}$

time taken to fill the $\frac{3}{45}$ part by inlet pipe

$= \frac{3}{45} \times 11.25 = \frac{3}{4}$ hours

$= \frac{3}{4} \times 60 = 45$ minute

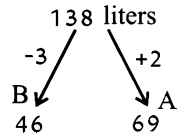
Total time = 42 hour 45 minutes

103. Pipe A is an inlet pipe which can fill an empty tank in 69 hours. Pipe B can empty a full filled tank in 46 hours. Starting with pipe B in the filled tank one by one, alternate both the pipes were open for one-one hour. How much time will it take to empty the tank?

(a) 11 day 12 hours (b) 11 day 10 hours
(c) 1 day 13 hours (d) 11 day 7 hours

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (d) Let total capacity



Emptied in 2 hours = 1 lit.
Emptied in (2 × 135) hours = 135 lit.
The last 3 liters will be emptied by pipe B in 1 hour.
Hence total time taken = 271 = 11 day 7 hours

104. An inlet pipe and an outlet pipes are opened in an order of one hour to fill and to empty a tank. When the tank is empty, starting is done from inlet pipe to fill the empty tank inlet pipe takes 10.5 hours, and the outlet can empty the full tank in 35 hours. How much time will be taken to fill in the tank?

(a) 28 hours 7 minutes
(b) 29 hours
(c) 30 hours
(d) 28 hours 42 minutes

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (d) Part filled by pipe A in 1 hour = $\frac{1}{10.5}$

Emptied part by pipe B in 1 hour = $\frac{1}{35}$

Filled part by (A – B) in total 2 hours = $\frac{1}{10.5} - \frac{1}{35}$

$= \frac{2}{21} - \frac{1}{35}$

$= \frac{70-21}{21\times 35} = \frac{49}{35\times 21} = \frac{7}{105} = \frac{1}{15}$

time taken to fill $\frac{1}{15}$ part = 2 hours

and filled part of tank in 2 × 14 hour. = $\frac{14}{15}$

Remaining part = $1 - \frac{14}{15} = \frac{1}{15}$

Now time taken to fill $\frac{1}{15}$ part by pipe A.

$= \frac{1}{15} \times 10.5 = 0.7$ hour

$= 60 \times 0.7 = 42$ minute

Hence total time taken to fill the tank = 28 hour 42 minute

105. Pipe A is an inlet pipe, which can fill a tank in 57 hours. Pipe B can empty the same tank in 38 hrs. When the tank is full, both pipes are opened alternately for one hour each time starting with A. How much total time will be taken to empty the tank?

(a) 9 day 13 hours (b) 9 day 10 hours
(c) 9 day 12 hours (d) 9 day 7 hours

RRB Group-D – 15/11/2018 (Shift-III)

Ans. :(c) Filled part by pipe A in 1 hour $=\frac{1}{57}$

emptied part by pipe B in 1 hour $=\frac{1}{38}$

emptied part by both pipes in 2 hours $=\frac{1}{38}-\frac{1}{57}$

$=\frac{57-38}{57\times38}=\frac{19}{57\times38}=\frac{1}{114}$ part

time taken to fill $\frac{1}{114}$ part = 2 hours.

∴ Total time taken to empty the entire tank 2 × 114 = 228 hours

228 hours = 9 day 12 hours

106. Pipe A can empty a filled tank in 32 hours, while B can fill this empty tank in 40 hours. If pipe A and B is used alternately for 1 hours each time when the tank is filled, then what time will be taken to empty the tank?

(a) 315 hours (b) 320 hours
(c) 319 hours (d) 311 hours

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (d) Filled part of the tank in 1 hour = $\frac{1}{40}$

emptied part of the tank in 1 hour = $\frac{1}{32}$

emptied part in 2 hours = $\frac{1}{32}-\frac{1}{40}=\frac{5-4}{160}=\frac{1}{160}$

Time taken to fill the $\frac{1}{160}\times155$ part of tank = 2 × 155

Remaining part $=\frac{5}{160}=\frac{1}{32}$

time take to empty $\frac{1}{32}$ part = 1 hour

Hence total time taken to empty the entire tank

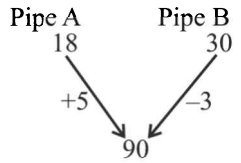
= 310 + 1 = 311 hours.

107. Pipe A could fill an empty cistern in 18 hours while Pipe B can drain a filled cistern in 30 hours. When the cistern is empty, Pipe A is turned on for an hour and then turned off. Now Pipe B is allowed to drain out water from the cistern for an hour and then turned off. The pipes were alternately left open for an hour each time till the cistern was full. How much time did it take for the cistern to be full?

(a) 45 hours
(b) 90 hours
(c) 86 hours 40 minutes
(d) 86 hours 48 minutes

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (d)



∴ work done by pipe A and B alternately in 2 hours = +5 – 3 = 2 unit

∴ 1 Cycle (2 hours)→ 2 unit

↓ × 43 ↓ × 43

86 hours 86 unit

time taken by pipe A to fill the reaming 4 units of work

$=\frac{4}{5}\times60=48$ minutes

Hence total time to fill the tank = 86 hour 48 min.

108. Two pipes, when working one at a time can fill a cistern in 2 hours and 3 hours, respectively while a third pipe can drain the cistern empty in 6 hours. All the three pipes were opened together when the cistern was $\frac{1}{6}$ full. How long will it take for the cistern to be completely full ?

(a) 1 hour 15 minutes
(b) 1 hour 30 minutes
(c) 1 hour 20 minutes
(d) 1 hour

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (a) All three cistern will be filled in one hour $=\frac{1}{2}+\frac{1}{3}-\frac{1}{6}$

$=\frac{3+2-1}{6}=\frac{4}{6}=\frac{2}{3}$ part

∴ Remaining part of cistern $=1-\frac{1}{6}=\frac{5}{6}$

∴ Time taken to fill remaining cistern $=\frac{\frac{5}{6}}{\frac{2}{3}}$

$=\frac{5}{6}\times\frac{3}{2}=\frac{5}{4}=1\frac{1}{4}$ hours

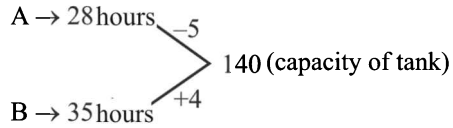
i.e. 1 hour 15 minutes

109. Pipe A can empty a filled tank in 28 hours while Pipe B can fill the same tank, when empty, in 35 hours. If alternately Pipes A and B are turned on for an hour each time, starting when the tank is full, how long will it take to empty the tank ?

(a) 279 hours (b) 271 hours
(c) 275 hours (d) 280 hours

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (b)



Emptied part by A in 1 hour = 1 × 5 = 5 units
Filled part by B in 1 hour = 1 × 4 = 4 units
∴ Tank emptied in 2 hours = – 5 + 4 = – 1 units
∵ 1 unit tank empties in 2 hours.
∴ 135 units will be emptied by tank (A + B)
= 135 × 2 = in 270 hours
And remaining 5 units will be emptied = In 1 hour (by A)
∴ Total time = 270 + 1 = 271 hours.

Type - 6

110. Pipe P can take out whole water from a tank in 20 hours. Another pipe Q can take out 20 liters per hours. If both the pipes are opened, then tank gets emptied in 12 hours, Find the capacity of the tank?

(a) 400 liters (b) 800 liters
(c) 650 liters (d) 600 liters

RRB JE - 02/06/2019 (Shift-II)

Ans. (d) $\frac{1}{12}-\frac{1}{20}=\frac{20-12}{240}=\frac{8}{240}\Rightarrow 30$ hours.
= 30 × 20 = 600 liters

111. There is a hole in a tank, which can empty it in 8 hours, one more pipe has been planted for the tank which can fill 6 litre of water per hour. pipe is opened, but due to (leakage) hole the tank gets empty in 12 hours. What is the capacity of the tank?

(a) 80 liters (b) 120 liters
(c) 144 liters (d) 78 liters

RRB JE - 27/06/2019 (Shift-I)

Ans : (c) 1 hour pure filler work of second tap

$$=\left(\frac{1}{8}-\frac{1}{12}\right)=\frac{1}{24}$$

It there were no other holes, then the tank would be filled in 24 hours.
∴ Tank capacitance = 6 × 24 = 144 liters.

112. If a bucket is filled 80% then it has two litre more water than when it is filled $66\frac{2}{3}\%$. Find the capacity of the bucket?

(a) 20 litre (b) 10 litre
(c) 15 litre (d) 12 litre

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) $80\% - 66\frac{2}{3}\% = 2$ liters

$$\frac{240-200}{3}\% = 2 \text{ liters}$$

$$\frac{40}{3}\% = 2 \text{ liters}$$

$$100\% = \frac{2\times 3}{40}\times 100 = 15 \text{ liters}$$

Hence capacity of Bucket = 15 liters

113. Kajipet, whose population is 4000, needs 9 litre water per day per person. It has tank of 15 m × 8m × 6m cuboid shaped. If the tank is fully filled with water, then the water of the tank will lest for how many days?

(a) 20 day (b) 25 day
(c) 10 day (d) 30 day

RRB Group-D – 19/09/2018 (Shift-I)

Ans : (a) Number of days $=\frac{\text{Volume of cuboid}}{\text{Population} \times \text{Necessity}}$

$=\frac{15\times 8\times 6\times 1000}{4000\times 9}$ (∵ 1 m^3 = 1000 liters)
= 20 days

114. 2/5 of a vessel is filled with oil. When 20 litres of oil is used then it is 7/12 full. Find the capacity of the vessel?

(a) 1200 litres (b) 1400 litres
(c) 1600 litres (d) 1000 litres

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (a) Part of oil present in the vessel = $\frac{3}{5}$ part
Part of left oil after extracting 20 liters of oil from the same vessel $=\frac{7}{12}$ part
Therefore, the part which was filled with 20 liters of oil

$$=\frac{3}{5}-\frac{7}{12}$$

$$=\frac{36-35}{60}=\frac{1}{60} \text{ part}$$

∵Quantity of oil present in the $\frac{1}{60}$ portion = 20 liters
∴ Quantity of oil present in one part

$$=\frac{20\times 60}{1}=1200 \text{ liters}$$

Hence the capacity of vessel is 1200.

115. A water container was 3/5 full, when 38 litres of water is taken out of it, then 1/8 part remains what is the total capacity of the container?

(a) 60 litres (b) 65 litres
(c) 75 litres (d) 80 litres

RRB NTPC 03.04.2016 Shift : 3

Ans : (d) Let the total efficiency of the container be x*l*.
According to the question,

$$\frac{3}{5}x - 38 = \frac{1}{8}x$$

$$\frac{3}{5}x - \frac{x}{8} = 38$$

$$\frac{24x - 5x}{40} = 38$$

$$19x = 38 \times 40$$

$$x = 2 \times 40 \qquad x = 80\ l$$

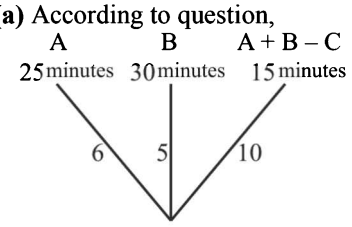
Hence total capacity of container is 80 *l*.

116. Two pipes namely A, B can fill a tank in 25 minutes and half an hour respectively and a pipe C can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is:

(a) 450 gallons (b) 300 gallons
(c) 240 gallons (d) 600 gallons

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (a) According to question,



L.C.M. of 25, 30, 15 = 150

or, A + B – C = 10

6 + 5 – C = 10

or C = 1

Hence time taken to empty the tank by C $= \frac{150}{1}$ min = 150 min

Thus the capacity of tank = 150 × 3 Gallon water ⇒ $\boxed{450 \text{ Gallon}}$

Type - 6

117. On starting pipe A, B and C together, they can fill any tank in 4 hours. Pipe a fills twice than that of pipe B and pipe C fills at the twice the speed of pipe A. How much time will by taken by pipe A to completely fill the tank?

(a) 7 hours (b) 30 hours
(c) 28 hours (d) 14 hours

RRB RPF SI – 12/01/2019 (Shift-II)

Ans. (d) : Time taken to filling the tank together by pipes = 4 hours

Let the time taken by pipe A to fill the tank = x hours

∴ The time taken by pipe B to fill the tank = 2x

∴ Time taken by pipe C to fill the tank $= \frac{x}{2}$

∴ Filled part of the tank by pipe A, B and C in 1 hour

$$\Rightarrow \frac{1}{x} + \frac{1}{2x} + \frac{1}{\frac{x}{2}} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{2x} + \frac{2}{x} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x}\left[1 + \frac{1}{2} + 2\right] = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x}\left[\frac{2+1+4}{2}\right] = \frac{1}{4}$$

$$\Rightarrow \frac{1}{x} \times \frac{7}{2} = \frac{1}{4}$$

$$\Rightarrow x = \frac{7 \times 4}{2}$$

$x = 7 \times 2 = 14$ hours

$\boxed{x = 14 \text{ hours}}$

Hence the time taken by pipe A to fill the tank alone = 14 hours.

118. A tank can be filled by two pipes in which one pipe can fill with the thrice speed as compared to other. If both the pipes together fill the tank in 36 minutes, then the slow pipe alone can fill the tank in how much time?

(a) 81 minutes (b) 144 minutes
(c) 120 minutes (d) 108 minutes

RRB JE - 30/05/2019 (Shift-III)

Ans : (b) Let a slow-speed pipe fills the tank in x min.

then the high-speed pipe will fill it in $\frac{x}{3}$ min.

$$\therefore \frac{1}{x} + \frac{3}{x} = \frac{1}{36}$$

$$\frac{4}{x} = \frac{1}{36}$$

x = 144

Hence slow-speed pipe will fill the tank in 144 minutes

119. Pipe P works twice as fast as Q. Q pipe works twice faster as R. All three pipes together can fill any tank in 8 hrs. How long Q alone takes to fill the tank?

(a) 56 hours (b) 14 hours
(c) 25 hours (d) 28 hours

RRB JE - 26/06/2019 (Shift-I)

Ans : (d) According to the considered efficiency, the time taken by P, Q, R pipes to fill the tank is 4x, 2x and x hour respectively.

According to question-

$$\frac{1}{4x} + \frac{1}{2x} + \frac{1}{x} = \frac{1}{8}$$

$$\frac{1+2+4}{4x} = \frac{1}{8}$$

$$\frac{7}{4x} = \frac{1}{8}$$

x = 14 hours

Hence time taken by Q to fill the tank = 2x

= 2 ×14 = 28 hours

120. Time taken by three taps P, Q and R individually to fill a tank is in the ratio of 2:1:6 which of the tap is fastest in these?

(a) can not be determined (b) P
(c) Q (d) R

RRB JE - 27/06/2019 (Shift-I)

Ans : (c)

	P		Q		R
Time	2	:	1	:	6
Efficiency	$\frac{1}{2}$	:	$\frac{1}{1}$	:	$\frac{1}{6}$

LCM of 2, 1, 6 = 6

$$\frac{1}{2}\times 6 : \frac{1}{1}\times 6 : \frac{1}{6}\times 6$$

$$3 : 6 : 1$$

Hence the capacity of the tap Q is high then this tap will be the fastest.

121. Three pipes P, Q and R can fill a tank in 30 min, 20 min and 10 min. When the tank is emptied, then the three pipes are opened while they discharge three chemical solutions, S, T and U respectively. Find the ratio of the solution U in the material present in the tank after 3 minutes?

(a) 7/11 (b) 4/11
(c) 6/11 (d) 5/11

RRB JE - 28/06/2019 (Shift-III)

Ans. (c) Filled part by P, Q and R in 1 minute

$$= \frac{1}{30}+\frac{1}{20}+\frac{1}{10}$$

$$\frac{2+3+6}{60}=\frac{11}{60}$$

Filled solution by R in 3 minutes = $\frac{1}{10}\times 3=\frac{3}{10}$

Ratio of solution U to R = $\frac{\frac{3}{10}}{\frac{11}{20}}=\frac{3\times 20}{11\times 10}=\frac{6}{11}$

122. If a pipe P fills a tank in 10 minutes. Five pipes, in which the capacity of each pipe is 20% of pipe P, can fill the tank in?

(a) 10 minutes (b) 50 minutes
(c) 130 minutes (d) 175 minutes

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (a) Efficiency of P = 10 minutes

$\therefore$ 1 min efficiency of P = $\frac{1}{10}$

20% efficiency of P = $\frac{1}{10}\times\frac{20}{100}=\frac{1}{50}$

time to fill the tank by 5 pipes

$$= \frac{1}{50}+\frac{1}{50}+\frac{1}{50}+\frac{1}{50}+\frac{1}{50}$$

$$= \frac{5}{50} = \frac{1}{10}$$

Hence total time = 10 minutes

123. P, Q and R are channels which discharge solutions A, B, C respectively in a tank. When the tank is empty and all the three channels are opened, what is the proportion of the solution C in the tank after 3 minutes, if the channels, P, Q and R can fill the tank in 30 minutes and 20 minutes and 10 minutes respectively when they are opened simultaneously?

(a) 6 (b) $\frac{5}{11}$
(c) 5 (d) $\frac{6}{11}$

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (d) The ratio of work done by P, Q and R in 3 minutes are as follows

$$=\frac{3}{30}:\frac{3}{20}:\frac{3}{10}=\frac{1}{3}:\frac{1}{2}:\frac{1}{1}=2:3:6$$

i.e. the ratio of P, Q and R = 2 : 3 : 6

then the ratio of the solution = $\frac{6}{2+3+6}=\frac{6}{11}$

124. Working with full speed a pipe can fill can empty tank in 2 hrs. Although, during first hour it worked at 1/12 of its efficiency, during second hour it worked at 1/9 of its efficiency, during third hour it worked at 1/6 of its efficiency, during fourth hour it worked at 1/4 of its efficiency and during fifth hour the efficiency was 1/3 to that compared to which it should work. The second pipe also worked like this, but if it worked at its full speed, then it would have filled the tank in 4 hrs, if both the pipes and that pipe which empties the tank at a fixed speed, starts together them half emptied tank can be filled in 5 hrs. If alone the pipes which empties the tank is started then how much hours will it take to empty the tank?

(a) 20 (b) 24
(c) 30 (d) 32

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (b) Let the emptying pipe empties the tank in x hour.

From question-

$$\frac{1}{2}\left[\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}\right]+\frac{1}{4}\left[\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}\right]-\frac{5}{x}=\frac{1}{2}$$

$$\frac{1}{2}\times\frac{34}{36}+\frac{1}{4}\times\frac{34}{36}-\frac{5}{x}=\frac{1}{2}$$

$$\frac{17}{36}+\frac{17}{72}-\frac{5}{x}=\frac{1}{2}$$

$$\frac{34+17}{72}-\frac{1}{2}=\frac{5}{x}$$

$$\frac{51}{72}-\frac{1}{2}=\frac{5}{x}$$

$\frac{51-36}{72}=\frac{5}{x}$

$\frac{15}{72}=\frac{5}{x}$

$x=\frac{5\times 72}{15}$

x = 24

Hence the emptying pipe will empty the tank in 24 hours.

125. One of the two water filling pipe in a tank works 1.5 times more efficiently than the other. If these two pipes works with such adraining pipe which alone can empty the tank in 12 hours, then the empty tank can be filled in 28 hours. How much time will be taken by the less efficient pipe to fill the tank?

(a) 21 (b) 18
(c) 24 (d) 15

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (a) Let working efficiency of tank A = 1
then, the working efficiency of second tank B = 1.5
A : B = 1 : 1.5 = 10 : 15 = 2 : 3
the ratio of times of A and B will be inversely proportional to their efficiencies.
hence ratio of times = 3 : 2
hence the time of A is 3x and time of B is 2x.
According to the question-

$\Rightarrow \frac{1}{2x}+\frac{1}{3x}-\frac{1}{12}=\frac{1}{28}$

$\Rightarrow \frac{5}{6x}=\frac{1}{28}+\frac{1}{12}$

$\Rightarrow \frac{5}{6x}=\frac{12+28}{28\times 12}$

$\Rightarrow \frac{5}{x}=\frac{40}{56}$ $\boxed{x=7}$

time taken to fill the tank with less efficient pipe = 3x
= 3 × 7 = 21 hours

126. A water tank is filled with the help of three pipes X, Y and Z in 5 hours. Pipe Z is three times faster than pipe Y and pipe Y is two times faster than pipe X. How much time will be taken by the pipe X alone to fill the tank?

(a) 35 hours (b) 40 hours
(c) 60 hours (d) 45 hours

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (d) Let the time taken to fill the tank by pipe = t hour
According to the question-

$\frac{1}{t}+\frac{2}{t}+\frac{6}{t}=\frac{1}{5}$

$\frac{9}{t}=\frac{1}{5}\Rightarrow \boxed{t=45\text{ hours}}$

Hence pipe x will fill the tank in 45 hours.

127. Two inlet pipe, in which the first inlet pipe is twice capable than the second pipe. One another outlet pipe which can empty the filled tank in 12.5 hours, working together can fill the empty tank in 2.5 hours. In how many hours the less capable inlet pipe will fill the tank?

(a) 7.5 (b) 6.25
(c) 5 (d) 8.75

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (b) Let there be three pipes A, B and C

A = 2 × B

$C=\frac{1}{12.5}h$ (Empties)

$A+B-C=\frac{1}{2.5}$

$A+B=\frac{1}{2.5}+\frac{1}{12.5}=\frac{10}{25}+\frac{10}{125}$

$A+B=\frac{50+10}{125}=\frac{60}{125}$

$\because A=2B$

$3B=\frac{60}{125}\Rightarrow B=\frac{20}{125}=\frac{4}{25}$

Hence the time taken to fill the less efficient pipe

$=\frac{25}{4}=6.25$ hours.

128. A pipe, working with its full speed can fill an empty tank in 2 hours. Although during its first two hours it worked with 1/12 part of its efficiency during its second two hours it worked with 1/9 part of its efficiency, during third two hours it worked with 1/6 part of its efficiency, during fourth two hours it worked with 1/4 part of its efficiency and during its fifth two hours it worked with 1/3 part of its efficiency. Another pipe also shows the same type of performance, but if it works at it full speed then it would have filled the tank in 4 hours. A drain pipe is also there which takes out the water from the tank at a fixed rate. If all the three pipes work together then the empty tank can be filled in 10 hrs. If rest of the two pipes remain closed then in how many hours the drain pipe will empty the full tank completely?

(a) 24 (b) 30
(c) 20 (d) 32

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (a) Let the emptying pipe empties the tank in total x hour.

According to the question-

$$\Rightarrow \frac{1}{2}\left[2\left\{\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}+\right\}\right]+\frac{1}{4}\left[2\left\{\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}\right\}\right]-\frac{10}{x}=1$$

$$\Rightarrow 2\left[\frac{3}{4}\left(\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}\right)\right]-\frac{10}{x}=1$$

$$\Rightarrow \frac{3}{4}\times\frac{34}{36}-\frac{5}{x}=\frac{1}{2}$$

$$\Rightarrow \frac{17}{24}-\frac{5}{x}=\frac{1}{2}$$

$$\Rightarrow \frac{5}{x}=\frac{17}{24}-\frac{12}{24}$$

$$\Rightarrow \frac{5}{x}=\frac{5}{24}$$

x = 24 hours

129. One inlet pipe among the two inlet pipes works twice as efficient as the other inlet pipe. A drain pipe can empty a fully filled tank in 12 hours. If all the three pipes are opened together, then the empty tank is filled in 12 hours. In how many hours the less efficient inlet pipe will alone fill the empty tank?

(a) 15 (b) 18

(c) 12 (d) 9

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (b) Let the another pipe fills the tank in x hours then first pipe will take $\left(\frac{x}{2}\right)$ hour (because works from double efficient). A third pipe empties the tank in 12 hours. Then According to the question,

$$\frac{1}{\frac{x}{2}}+\frac{1}{x}-\frac{1}{12}=\frac{1}{12} \text{ or}$$

$$\frac{2}{x}+\frac{1}{x}-\frac{1}{12}=\frac{1}{12}$$

$$\Rightarrow \frac{3}{x}=\frac{1}{12}+\frac{1}{12}$$

$$\Rightarrow \frac{3}{x}=\frac{2}{12}$$

$\Rightarrow x = 18$ hours.

130. A tank is filled by three pipes X, Y and Z in 6 hours. If Z is three time faster than Y and Y is two time faster than X, then how many hours is needed by X to fill the tank?

(a) 27 hours (b) 54 hours

(c) 30 hours (d) 45 hours

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) The time taken by X, Y and Z is 6T, 3T and 2T respectively.

X : Y : Z

x : 2x : 6x → efficiency

6 : 3 : 1 → time

According to the question-

$$\frac{1}{6T}+\frac{1}{3T}+\frac{1}{T}=\frac{1}{6}$$

$$\Rightarrow \frac{1+2+6}{6T}=\frac{1}{6}$$

$$\Rightarrow T=9$$

Hence time taken to fill the tank by X = 6T

$= 6\times 9 = 54$ hours

131. Two inlet taps, in which one tap's capacity is 4.5 times greater than the other, an outlet tap which can empty a tank in 9 hours, working together can fill an empty tank in 7.5 hours. How many hours will be taken by the less efficient pipe to fill the tank?

(a) 18 (b) 27

(c) 22.5 (d) 13.5

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (c) : If pipe A fills a tank in x hours then pipe B will fill that tank in $\frac{x}{4.5}$ hours because the efficiency of B is 4.5 times more than A.

According to question-

$$\left(\frac{1}{x}+\frac{4.5}{x}\right)-\frac{1}{9}=\frac{1}{7.5}$$

$$\frac{5.5}{x}-\frac{1}{9}=\frac{2}{15}$$

$$\frac{49.5-x}{9x}=\frac{2}{15}$$

$\Rightarrow 5(49.5 - x) = 6x$

$\Rightarrow 247.5 - 5x = 6x$

$\Rightarrow 11x = 247.5$

$$\Rightarrow x=\frac{2475}{110}$$

$\Rightarrow x = 22.5$ hours

132. Among the two inlet pipes A and B, A's efficiency is twice than that of B. Pipe C is an outlet pipe which takes three times the time taken by pipe A in emptying the filled reservoir. If the three pipes are opened together when the reservoir is empty then it takes 6 hours to fill the reservoir. When A and C are working together, then how much time will be taken to fill the reservoir?

(a) 15.75 hours (b) 10.5 hours
(c) 21 hours (d) 42 hours

Ans. (b) : Let the time taken by A to fill the tank is x hours.

The time taken by B to fill the tank will be 2x hours

The time taken by C to empty the tank will be 3x hours

According to question-

$$\frac{1}{x}+\frac{1}{2x}+\frac{1}{3x}=\frac{1}{6} \quad =\frac{6+3-2}{6x}=\frac{1}{6}$$

x = 7 hours

A = 7 hours, B = 14 hours, C = 21 hours

Filled part of the tank by A and C together in 1 hour =

$$\frac{1}{7}-\frac{1}{21}=\frac{2}{21}$$

Hence time taken to fill the reservoir by A and C together $=\frac{21}{2}=10.5$ hours

133. A tap can fill a water tank in 4 hours. The tap was opened and 3/4 part of the tank was filled then two taps of same filling capacity were opened. Then after 15 minutes, another tap of a capacity of two times the first tap was opened in order to take out the water from the tank. Calculate the time which it takes to fill the tank.

(a) 4 hours 15 minutes
(b) 4 hours
(c) 3 hours
(d) 3 hours 30 minutes

RRB Group-D – 15/11/2018 (Shift-I)

Ans : (d) According to question, total filled part of the tank by pipes in 3 hour 15 min.

$$=\frac{3}{4}+\frac{1}{16}+\frac{1}{16}+\frac{1}{16}=\frac{15}{16}$$

let the time taken to fill the remaining 1/16 part of the tank is x hours

$$\therefore \quad \frac{x}{4}+\frac{x}{4}+\frac{x}{4}-\frac{x}{2}=\frac{1}{16}$$

$$\frac{3x}{4}-\frac{2x}{4}=\frac{1}{16}$$

$$\frac{x}{4}=\frac{1}{16} \Rightarrow x = 15 \text{ minutes}$$

time taken to fill the entire tank

= 3h. 15m + 15 min = 3 hour 30 min.

134. In two inlet pipes, one is twice capable than the other. There is also a third outlet pipe which can empty the fully filled tank in 10 hours, it takes 2.5 hours to fill the tank completely. How many hours will be taken by the low capacity pipe to fill the tank alone?

(a) 6 (b) 7
(c) 8 (d) 5

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (a) : Let,

Time taken by A = 2n

Time taken by B = n

from question-

$$\frac{1}{2n}+\frac{1}{n}-\frac{1}{10}=\frac{10}{25}$$

$$\frac{1}{2n}+\frac{1}{n}=\frac{2}{5}+\frac{1}{10}$$

$$\frac{1+2}{2n}=\frac{4+1}{10}$$

$$\frac{3}{2n}=\frac{5}{10}$$

$$\boxed{n = 3}$$

time taken by less efficient pipe = 2n = 2 × 3 = 6 hours.

135. Pipe A fills the tank in 1/4 time of B. Pipe C takes three times more time as compared to A. If all three pipes are opened together, they can fill an empty tank in 24 hours. If pipe C was not opened then how much time will be taken to fill the empty tank?

(a) 30.2 (b) 28.4
(c) 30.6 (d) 30.4

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (d) : Given-

A : B = 1 : 4

A : C = 1 : 3

A : B : C = 1 : 4 : 3

Let A, B, C fill the tank in x, 4x and 3x hours respectively.

According to question-

the ratio of efficiencies= A : B : C = $1:\frac{1}{4}:\frac{1}{3}$

= 12 : 3 : 4

total work = 19 × 24 = 456

total time taken by (A + B) = $\frac{456}{15}$ = 30.4 hours

136. One of the two inlet pipes works twice as efficient as the other. A drain pipe that can empty a tank in 8 hours, can fill the empty system is 8 hours. How many hours will the less efficient inlet pipe take to fill the empty system by itself ?

(a) 12 (b) 6

(c) 10 (d) 8

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (a) Let the high efficiency pipe fills the tank in x hours.

A low efficiency pipe fills the tank in 2x hours.

and the third empties that tank in 8 hours.

According to question-

$$\frac{1}{x}+\frac{1}{2x}-\frac{1}{8}=\frac{1}{8}$$

$$\frac{3}{2x}=\frac{2}{8}$$

x = 6 hours

Hence the time to fill the tank with low efficient pipe alone = 2x = 12 hours

137. A pipe, working at full speed, can fill an empty cistern in 1 hour. However, during the first hour it worked at one-twelth of its capacity, during the second hour at one-ninth of its capacity, during the third hour at one-sixth of its usual capacity, during the fourth hour at one-fourth of its usual capacity and during the fifth hour it was only one-third as efficient as it was supposed to be. A second pipe also displayed similar performance, but it worked at full speed would have filled the empty cistern in 2 hours. Together with a drain pipe that drained water out of the tank at a constant rate, the empty cistern could be filled in 5 hours, all the three pipes working simultaneously. How many hours will it take the drain pipe to empty the filled cistern if no other pipe was functioning during the time?

(a) 16 (b) 12

(c) 10 (d) 15

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (b) Let drain pipe will be emptied the cistern in x hours

filled part of cistern in 5 hours by first pipe

$$=\frac{1}{12}+\frac{1}{9}+\frac{1}{6}+\frac{1}{4}+\frac{1}{3}$$

$$=\frac{3+4+6+9+12}{36}=\frac{34}{36}$$

then filled part of cistern in 5 hours by second pipe

$$=\frac{1}{2}\times\frac{34}{36}=\frac{17}{36}$$

when all three pipes are opened simultaneously-

$$\frac{34}{36}+\frac{17}{36}-\frac{5}{x}=1$$

$$\frac{51}{36}-1=\frac{5}{x},$$

$$\frac{15}{36}=\frac{5}{x},\quad x = 12 \text{ hours.}$$

138. Sita's bucket has thrice the capacity as Ramu's bucket. It takes 60 turns for Sita's bucket to fill the empty drum. How many turns it will take for both the buckets of Sita's and Ramu's, having each turn together to fill the empty drum?

(a) 45 (b) 40

(c) 50 (d) 30

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (a) Let the capacity of Ramu's bucket = x

then capacity of Sita's bucket = 3x

capacity of drum = 60 × 3x

= 180 x

let both fill the bucket N times and turn into the drum

∴ N (3x+x) = 180x

$$N=\frac{180x}{4x}=45$$